AOBP Platform

Architecture, API and Engineering Specification

Secure browser access with BFF and corporate SSO

AWS EKS · Application Load Balancer · Spring Cloud Gateway · Spring Boot BFF

Selected browser business route

ALB → API Gateway → BFF → private business APIs

Version 2.3 | 2 November 2026

Retains the source document's 23-section structure. Includes plain-language request journeys, implementation responsibilities, security risks, acceptance criteria and seven high-resolution architecture illustrations.

Design status: architecture baseline for implementation and validation. Proposed values are distinguished from standards and measured operating results.

Document guide

The numbered sections retain the source organization. Diagram pages use a larger landscape sheet so their text remains readable when zoomed or printed.

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1. Design scope and delivery status

Introduction

AOBP brings separately deployed React microfrontends together as one browser application. Users sign in through BAC corporate Single Sign-On (SSO), then work with flight information, workflows and other business services. This specification explains who verifies the user, where credentials are kept, and the exact route taken by each type of request.

The selected browser architecture is **AWS Application Load Balancer (ALB) → Spring Cloud API Gateway → Spring Boot Backend for Frontend (BFF) → private microservices**. The BFF runs inside EKS, behind the gateway. It manages the browser session and all user OAuth tokens. Each business service validates its incoming access token and enforces its own business permissions and data restrictions.

Think of the ALB as the building entrance, the gateway as traffic reception, the BFF as the user's secure representative, and a microservice as the department that makes the final business decision. Reception can direct a visitor, but only the department can decide whether that visitor may change a particular record.

Reading this specification

Start with sections 1-3 for the decision and the overall picture. Section 5 answers which URL goes where. Section 6 explains login, normal use, token renewal and logout in sequence. Section 13 shows attack points and their controls. Section 18 explains the AWS and Kubernetes deployment. The remaining sections preserve the platform engineering contracts from the source document.

Revision note: The current source already contains a BFF proposal. This revision retains all 23 main sections, clarifies that proposal, replaces the five referenced illustrations with newly drawn PNGs, and adds a machine-integration view and an attack-point view. It does not claim that the reviewer inspected a live deployment or that any implementation has been security-tested.

How to read this document

Start with the introduction and Figure 1 for the overall design. Follow Figure 2 for signing in and Figure 3 for a normal business action. Figure 4 explains why several BFF instances can safely serve the same user. Figure 5 shows where the components run in AWS. The later sections give developers and operators the precise contracts needed to implement that simple story.

Reader	Suggested route	What you will understand
Business or non-technical reviewer	Sections 1-3, Figures 1-7, decision table in 22	Why this design is chosen and how a user reaches protected data
Application developer	Sections 4-14	URLs, headers, session handling, API authorization and error contracts
Platform or operations engineer	Sections 16-20	EKS placement, networking, availability, deployment and recovery
Security reviewer	Sections 2, 5-8, 13, 21-23	Credential boundaries, threat controls and evidence needed for acceptance

The architecture in one minute

The user opens one AOBP web address. React draws the screens. For a business action, React calls the BFF through the load balancer and gateway. The BFF checks the user's server-side session, obtains a suitable access token and calls the private API. That API checks both the token and the business permission before reading or changing data. The result returns through the same components in reverse order.

Decision: place the BFF inside private EKS application subnets, behind the Spring Cloud API Gateway. Place the gateway behind the AWS ALB. Keep both as separate Kubernetes Deployments. This ordering is an AOBP design choice; OAuth standards define security responsibilities rather than requiring this exact gateway position.

Everyday question	Clear answer	Detailed section
Who sees the password?	BAC corporate identity provider; AOBP never collects it	6
Who keeps the API tokens?	BFF, in protected shared server-side storage	6
What does React keep?	An opaque session cookie managed by the browser; no OAuth tokens	4, 6
Who is contacted first?	ALB first; gateway first for dynamic application requests	3, 5
Does every request pass through BFF?	Browser business/session requests do; static assets and machine APIs have distinct routes	3, 5, 9
Can a service trust the gateway alone?	No. Every API validates its token and business permissions	7
What if a BFF pod fails?	Another replica reads shared state; live sockets reconnect	6, 8
What if a token expires?	BFF coordinates refresh; services never accept expired tokens as a fallback	6

How this resolves the reviewer's access-token concern

An access token is like an API access pass. A signed JWT is normally readable by whoever possesses it; signing does not encrypt its contents. A bearer token can be used by its holder until it expires or an enforced revocation rule rejects it. Giving that pass to browser JavaScript increases the places it can be copied from. Here the browser instead receives a session reference: the BFF keeps the API pass on the server and attaches it only to approved private API calls.

This reduces token theft from JavaScript storage and centralizes refresh/logout handling. It still requires strong browser security, because malicious page code can issue requests using the user's session. HttpOnly cookies, CSRF protection, CSP, controlled microfrontend releases and service authorization work together.

Security review concerns and design resolution

Concern to resolve	Resolution in this specification
Browser-managed OAuth would expose API credentials to JavaScript	Register the BFF as the confidential client; React receives no access, refresh or ID token
The Markdown already describes a BFF but leaves the edge generic	Specify AWS ALB, Kubernetes routing, gateway placement and private service paths
Authentication and authorization responsibilities can be confused	Corporate IdP authenticates; BFF manages sessions; microservices enforce business authorization
Session sharing does not fully specify token sharing	Persist both session and authorized-client state; coordinate concurrent refresh across BFF replicas
Gateway operation checks might be read as sufficient security	Gateway checks are coarse; service RBAC, ABAC and transaction rules remain mandatory

A token is a digital credential, not an encryption mechanism. TLS protects transport. A BFF reduces exposure of bearer tokens to browser JavaScript; it does not eliminate XSS, stolen-session or backend compromise risks. The reviewed Markdown already moves in the right direction. This revision makes its security boundary and implementation obligations explicit.

Scope and interpretation

The reviewed 23-section structure, engineering tables and step-by-step explanations are retained. Sections 3, 5, 6 and 18 make the placement and EKS topology concrete. Any earlier SPA-owned token flow is superseded for this web application. Business roles, group mapping and resource-scope concepts remain applicable.

AWS EKS is the reference deployment described here. This does not change the wider AOBP working direction of on-premises primary and AWS secondary/DR. Section 20 explains how the same application boundaries carry across sites. Whether EKS is initially a production or recovery runtime is an environment decision.

This is an implementation specification and architectural decision, not evidence of an installed system. Example domains, resource budgets, timeouts and lifetimes need environment configuration and validation. No load test, running repository or deployed cluster was inspected.

Plain-language glossary

Term	Meaning in this document
Authentication / SSO	Proving who the user is through the corporate sign-in system
Authorization	Deciding what that identified user may do with a particular record
BFF	A server acting for the browser; it keeps API tokens and manages the browser session
Access token / JWT	A signed API credential; its claims must be validated, not merely decoded
Refresh token	A server-side credential used to obtain another access token
ALB / gateway	ALB distributes incoming traffic; the application gateway adds route and traffic controls
EKS / Kubernetes	AWS-managed Kubernetes; Kubernetes schedules and operates application containers
Pod / replica	One running instance of a workload; replicas provide multiple instances
Ingress / Service	Ingress declares entry rules; a Service provides a stable Kubernetes endpoint for pods
Redis	A fast shared store used here for sensitive, temporary session and token state
TLS / mTLS	Encrypted transport; mutual TLS also authenticates both communicating workloads
CSRF / XSS	Forged browser requests / malicious code running in a page; they need different defenses
RBAC / ABAC	Permissions from business roles / decisions using trusted user and resource attributes
PKCE	A login-code protection mechanism binding the exchange to the initiating OAuth client
RTO / RPO	Target recovery time / acceptable data-loss window; both require business approval and tests

2. Architecture principles

Principle	What it means in everyday language	Implementation consequence
Secure by default	Start with access closed	Explicit routes, permissions and network allowlists; deny everything else
Identity-centric access	Trust the approved identity system	Corporate SSO through Keycloak; no application passwords
Keep credentials server-side	The browser does not carry API credentials	BFF stores tokens; browser holds an opaque HttpOnly session cookie
Defense in depth	One failed check must not open everything	WAF, gateway, BFF, service token checks and resource authorization
Least privilege	Give each person and workload only its required access	Scoped permissions, audiences, IAM roles and database grants
Backend authority	A hidden button is not security	Service checks action, object, fields and current business state
Separation of concerns	Each component has one clear job	ALB load-balances; gateway protects/routes; BFF manages sessions; services own business rules
Independent deployment	Change a UI module without rebuilding the security layer	Separate shell, remote, gateway, BFF and service deployments
Stable contracts	Deployment changes should not break browser URLs	Same-origin relative BFF URLs and documented OpenAPI contracts
Failure containment	A dependency problem must remain bounded	Deadlines, controlled errors, bulkheads and no silent authentication bypass
Traceability and privacy	Explain what happened without leaking secrets	Correlated logs, durable audit, redaction and restricted telemetry
Build once, promote unchanged	Test the same release that will run	Signed immutable image digests and reviewed GitOps promotion
Recoverability	Recovery needs a tested process	Backups, writer fencing, identity recovery and measured DR exercises

Why this design was chosen

Keeping tokens in the BFF gives one controlled place to refresh, revoke and protect them. Separating the gateway from the BFF lets traffic controls evolve without making the gateway the owner of browser identity. A common external origin keeps React integration simple even though frontend and backend deployments are separate. Service-side checks protect business data even if a route or UI control is wrong.

The cost is an additional application hop, a sensitive BFF runtime and a highly available session store. These are deliberate trade-offs. Capacity planning, patching and outage tests are required; the pattern alone is not a production-readiness certificate.

3. Logical architecture

Architecture overview

Figure 1. AOBP secure access architecture
See the landscape diagram plate following this section.

Read the diagram from top to bottom. Static UI files take the left path. Every browser business request takes the gateway and BFF path. The identity branch is used during login, refresh and logout; it is not a requirement to call the identity provider for every valid JWT.

Component	Placement	Responsibility
React shell and microfrontends	Independent static workloads in EKS	Draw screens; call relative BFF URLs; display safe session/capability information
AWS WAF and ALB	Regional entry; access scope matched to audience	WAF request rules, HTTPS termination, host/path rules and healthy-target distribution
AWS Load Balancer Controller	EKS infrastructure namespace	Reconciles Kubernetes Ingress into AWS resources; does not proxy application requests
Spring Cloud Gateway	Private EKS pods, aobp-edge	Request limits, allowlisted routes, tracing and partner-token checks
Spring Boot BFF	Private EKS pods, aobp-apps	OIDC client, login callback, CSRF, session, token refresh and relay
Shared session/token store	Private managed Redis-compatible service	Sensitive transient session and authorized-client state for all BFF replicas
Keycloak identity broker	Isolated identity runtime and database	Federates BAC SSO and issues AOBP tokens; maintains required federation metadata
Central Security Service	Private authorization runtime	Resolves approved group-to-role mappings, permissions and policy decisions
Flight / Workflow / other APIs	Private EKS pods	Validate JWTs and enforce business permissions, ABAC and state changes
Business data and event services	Private managed data tier	Durable records, audit outbox and governed events

Picture key and trust boundaries

Blue means browser and entry routing; teal means the BFF session boundary; green means business services; purple means identity; sand means sensitive shared state. Solid arrows show requests or credential exchanges. Dotted arrows in the EKS view show configuration rather than application traffic. Numbered arrows show the order for that particular journey, not a global ordering across every figure.

The gateway is permitted to forward a browser cookie to the BFF. It does not own that session or receive the BFF's outbound user tokens in this topology. The BFF creates a separate authenticated request to the business service. All components that handle cookies or tokens remain security-sensitive.

Exactly where the gateway and BFF sit

The **ALB is the first network entry**, and the **gateway is the first application component for dynamic AOBP traffic**. The BFF is behind it. The BFF is a separate Deployment and ClusterIP Service, not a component inside the React container or the gateway process. Start with one logical AOBP web BFF with multiple replicas, not one BFF per microservice. Split by distinct browser product or trust boundary only when there is a demonstrated need.

The reference gateway is Spring Cloud Gateway running in EKS. Amazon API Gateway is not an additional required hop. There is no extra Nginx ingress-controller data-plane hop in this design: the AWS Load Balancer Controller provisions an ALB that targets pods directly using IP targets. Kubernetes Services and EndpointSlices identify eligible pods; they are not always a physical extra hop on ALB IP-target traffic.

BFF-to-service requests use private Kubernetes service DNS over authenticated TLS. They do not leave the cluster and re-enter through the public ALB or gateway. A future internal east-west gateway is a separate decision; it is not needed for this reference flow.

Placement decision and alternatives

There are two sensible places for an application gateway: before the BFF, to govern incoming traffic, or after it, to govern API traffic. AOBP selects the first because the shared gateway already has an edge role and must protect login, session, WebSocket and partner entry paths. OAuth does not mandate this ordering. The following comparison makes the local decision reviewable.

Candidate	Benefit	Cost or constraint	AOBP decision
ALB → gateway → BFF → private APIs	One governed dynamic entry before the session handler; fits browser and machine routes	Additional hop; gateway must preserve OIDC, cookies and WS correctly	Selected
ALB → BFF → internal gateway → APIs	Central internal API routing and policy for BFF calls	Separate edge protections still needed; couples BFF requests to internal gateway availability	Valid alternative if internal API governance becomes a requirement
ALB → edge gateway → BFF → internal gateway → APIs	Additional independent API governance if explicitly required	Additional hop, latency and operational dependencies	Deferred until justified
ALB → combined gateway/BFF → APIs	Fewer deployments and hops	Session credentials and edge changes share one failure/release boundary	Not selected for this shared platform
ALB → gateway → APIs with browser-held tokens	Simple direct API access	Does not meet the requested server-side OAuth-token boundary	Superseded for the AOBP browser application

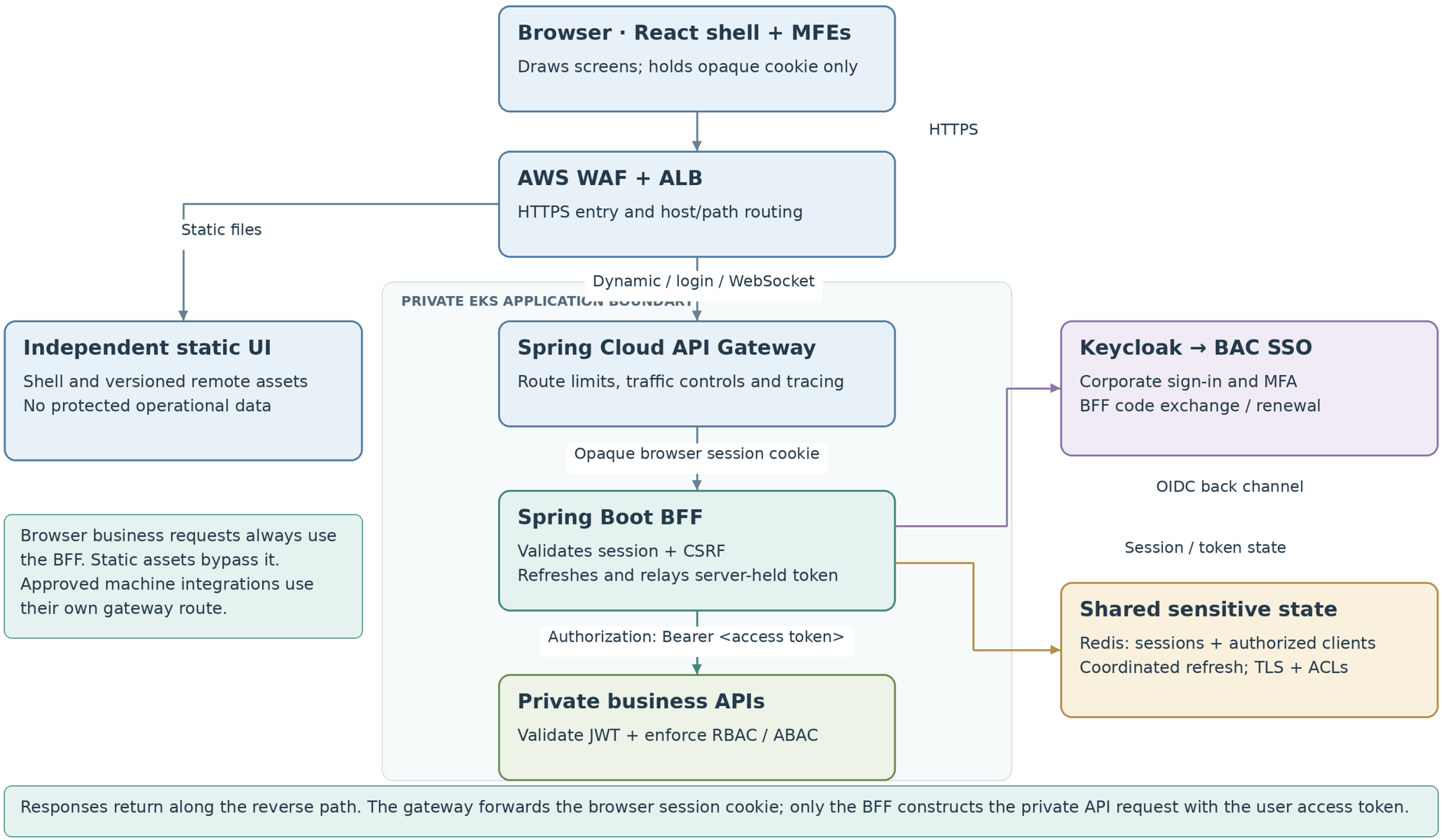
The gateway before the BFF does not validate a bearer token on browser routes because the browser has none. It forwards the session cookie to the BFF, which validates it. On machine routes, the gateway does validate the machine bearer token. These are separate security chains, even when hosted by the same gateway Deployment.

Three entry classes

1. **Static UI:** ALB → independently deployed shell or remote asset workload. No protected business data is embedded in public assets.
2. **Browser operations:** ALB → gateway → BFF → private business API. The same rule includes session discovery, login initiation, callbacks, logout and WebSocket upgrades.
3. **Approved system integration:** ALB → partner gateway route → private API, using a machine access token. It does not use a human BFF session.

Keycloak uses `identity.example.com` with independently managed authentication ingress. Its administrative and management endpoints remain private. Do not route Keycloak through a BFF that needs Keycloak to establish its own session. Identity cookies stay on the identity hostname; AOBP session cookies stay on the application hostname.

Figure 1. AOBP secure access architecture



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Illustrative topology · proposed controls

4. Microfrontend architecture

Reference composition: React shell plus independently published flight and workflow remotes, using Module Federation or an equivalent runtime loader. Confirm the bundler and federation implementation before selecting packages. Route-based independently deployed applications are an alternative if runtime federation complexity is not justified.

Item	Ownership and contract
Shell	Navigation, layout, authentication state UX, design-system bootstrap, error boundaries, remote loader
Flight remote	Flight views and interactions; calls `/bff/api/v1/flights/**`
Workflow remote	Workflow/task views; calls its approved BFF contract
Shared browser SDK	Typed API client, CSRF handling, trace propagation, sanitized error reporting
Shared design system	Versioned UI contracts; defined compatible ranges
Session	One BFF session; no tokens stored by individual MFEs
Remote manifest	Maps remote name/version to immutable asset URL and compatible shell contract

Example external paths: `/`, `/mfes/flight/1.4.0/remoteEntry.js`, `/mfes/workflow/2.1.0/remoteEntry.js`, `/bff/**`. The ingress sends MFE prefixes to their own static services; the shell fallback must not swallow `/bff`, `/mfes`, OIDC callbacks or health endpoints.

Release process: publish immutable remote assets, run compatibility tests, update the environment remote manifest, then invalidate only the short-lived manifest cache. Hash-named assets receive long immutable caching; the remote manifest receives short caching/revalidation. Keep previous remote versions during rollback and existing browser sessions. Avoid overwriting one `remoteEntry.js` URL for every release.

Share React through a tested compatible version contract; do not force every remote dependency into a global singleton. Use per-remote error boundaries so a failed workflow remote does not blank the flight UI. Add a remote-loading availability metric and a controlled reload prompt for incompatible sessions.

Same-page MFEs share JavaScript execution privileges. They are not security isolation boundaries. Restrict who may publish remote code, protect asset integrity, review dependencies, and apply CSP. Use a sandboxed separate-origin iframe only when actual isolation is required, with its own integration constraints. Do not promise SRI support without validating the selected remote loader.

5. Gateway-neutral contract and Spring reference

External routing contract

All browser examples use `https://aobp.example.com`. Frontend images are deployed separately, but appear under this one origin through ALB routing. Separate deployment does not require separate browser origins or broad CORS.

Host and path	ALB destination	Gateway destination	Security owner
aobp.example.com / and shell assets	UI static target	Not traversed	Asset pipeline; no operational secrets
aobp.example.com /mfes/**	Independent remote static target	Not traversed	Immutable remote assets and manifest governance
aobp.example.com /bff/api/**	Gateway	BFF REST	BFF session/CSRF; downstream resource authorization
aobp.example.com /bff/ws/v1	Gateway	BFF WebSocket endpoint	Session, Origin and authorized messages/subscriptions
aobp.example.com /oauth2/authorization/keycloak	Gateway	BFF login initiation	BFF creates state, nonce and PKCE transaction
aobp.example.com /login/oauth2/code/keycloak	Gateway	BFF callback	BFF checks correlation and exchanges code
aobp.example.com /bff/logout	Gateway	BFF logout	CSRF-protected POST
aobp.example.com /bff/oidc/backchannel-logout	Gateway; IdP-restricted where feasible	BFF logout-token receiver	Signed OIDC logout token; no browser session requirement
aobp.example.com /api/**	Reject	No browser bypass route	Gateway/edge deny
api.example.com /api/v1/flights/**	Partner gateway target	Flight API	Client credentials token; gateway + API validation
identity.example.com approved OIDC routes	Identity ingress	Not through AOBP gateway/BFF	Keycloak and corporate IdP
Any public /actuator/** or admin route	Reject	No public management route	Private operator network and authorization

Route priority must put known static/MFE and dynamic paths before the shell fallback. Unknown dynamic paths return an error; they must not turn into an HTML 200. The browser is not allowed to switch to partner APIs when its BFF session fails. Partner routes require machine credentials with explicit

audience, scopes and client controls; a hostname or CORS policy alone does not distinguish a trusted caller.

Request routing in everyday language

What the person or system does	Request journey	What proves identity
Opens the home page	Browser → ALB → static UI workload → browser	No business authentication needed to download public application code
Clicks Sign in	Browser → ALB → gateway → BFF; browser then follows redirects to Keycloak and corporate SSO	Corporate SSO authenticates the person; BFF validates the returned login transaction
Returns from SSO	Browser → ALB → gateway → BFF callback; BFF → Keycloak token endpoint	One-use code, state, PKCE, client authentication and OIDC validation
Reads or changes a business record	Browser → ALB → gateway → BFF → API; response reverses this route	Cookie to BFF; access-token Authorization header to API; API business permissions
Receives live updates	Browser → ALB → gateway → BFF WebSocket; authorized backend events → BFF → browser	Session at handshake plus ongoing subscription/delivery authorization
Logs out	Browser POST → ALB → gateway → BFF; BFF invalidates shared state and coordinates identity logout	Existing session and CSRF token; logout scope is explicit
Partner system reads an approved API	Partner → ALB → partner gateway route → API	Dedicated machine token; no BFF cookie or human refresh token

Only the callback delivers a temporary authorization code through the browser. API access tokens, refresh tokens and ID tokens stay server-side. Each returned API response contains business data, not these credentials.

Gateway responsibilities

The gateway enforces route/method allowlists, body/header limits, deadlines, request correlation and applicable quotas. It proxies BFF session cookies but does not interpret them as JWTs. Login and callback endpoints are forwarded without requiring an already authenticated session. The BFF still validates those endpoints; 'reachable before login' does not mean unvalidated.

Partner routes validate bearer tokens and coarse scopes before forwarding. BFF routes do not require a browser bearer token. Do not enable OAuth login, token storage or ALB OIDC authentication as a second session owner alongside the BFF. Avoid overlapping redirection or logout rules.

Strip browser-supplied identity headers such as X-User and X-Roles. Do not accept a browser Authorization header as a way to replace the token associated with the session. Sanitize forwarded-host/proto/client-IP information at trusted proxy boundaries. The BFF uses the configured external callback URL, not arbitrary request headers, to construct redirects. Preserve the external path through both proxies and test encoded-path and trailing-slash behavior.

Gateway access logs omit cookies, bearer headers and sensitive query strings, including OIDC code/state. Gateway quotas before session validation may be coarse network/route protections.

Per-user quotas belong after trustworthy user identification; do not key them on an unvalidated cookie or client-provided username.

Explicit BFF destination mapping

The BFF is an allowlisted application adapter, not an open proxy. The caller cannot provide a target URL, host, port or redirect destination. Preserve business query parameters only after validation. Never let a downstream redirect carry an access token to another host; disable automatic cross-host redirect following for credentialed calls.

External browser contract	Fixed private destination	Credential on the private call
GET /bff/api/v1/flights	Flight API: /api/v1/flights	Server-held access token in Authorization header
PATCH /bff/api/v1/flights /{id}/operational-times	Flight API: same resource under /api/v1	Server-held access token; validated If-Match
GET /bff/api/v1/tasks	Workflow API: /api/v1/tasks	Server-held access token for the approved audience
GET /bff/api/v1/session	Handled by BFF	No downstream API token returned
POST /bff/logout	Handled by BFF; approved IdP back channel if configured	BFF-held credentials only

Apply object-ID and path validation before building an outbound URI. Never concatenate an unvalidated suffix into a destination. Rate limits after session validation use a trusted subject/session reference; pre-login limits use bounded network/route controls.

Spring and replacement contract

Use Gateway Server WebFlux with Actuator, Micrometer and OAuth2 Resource Server for partner routes. Keep blocking database or policy calls out of reactive event-loop threads. BFF uses a separate Spring Boot application with OAuth2 Client and Spring Session. Pin a supported Spring Boot/Spring Cloud BOM combination during implementation; examples are not a request to reuse legacy versions.

Prefer explicit routes with stable IDs. Place WebSocket matching ahead of overlapping REST matching; use the selected release's supported WebSocket URI scheme and preserve upgrade headers. No business-resource policy belongs only in gateway configuration.

A replacement gateway must pass tests for callback paths, cookie attributes, CSRF forwarding, WebSocket upgrades, body limits, error contracts and trace context. Grafana provides monitoring; route changes use reviewed configuration and deployment. Gateway management endpoints stay private and read-only for routine monitoring.

6. Authentication and token lifecycle

What each credential means

Item	Purpose	Location	May React read it?
Corporate password / MFA evidence	Proves identity to BAC SSO	Corporate IdP interaction	Never collected by AOBP
Authorization code	Short-lived, one-use login exchange	Browser redirect to BFF callback	Appears transiently in callback URL; never used as an API token
ID token	Tells the BFF about completed authentication	BFF authorized-client/session state	No
Access token	Authorizes calls to the protected API audience	BFF store; Authorization header on private calls	No
Refresh token	Obtains a new access token	BFF server-side authorized-client store	No
Session cookie	Opaque reference to the BFF session	Browser cookie jar; BFF session lookup	No: HttpOnly
CSRF token	Proves a state-changing request came through the intended browser session	BFF session and safe CSRF response	Yes; it is not an API credential

The session cookie is still a sensitive credential. HttpOnly prevents normal JavaScript reads but does not stop malicious JavaScript from making requests through an already logged-in browser. CSP, dependency security, output encoding and controlled MFE publishing remain necessary.

Step 1 - Register the BFF for corporate SSO

Create one confidential OIDC client, for example `aobp-web-bff`, in Keycloak. Register exactly `https://aobp.example.com/login/oauth2/code/keycloak`, with exact approved logout redirects. Use Authorization Code with PKCE S256 plus confidential-client authentication. Prefer managed asymmetric client credentials where supported; otherwise keep a rotated client secret in AWS Secrets Manager with narrowly scoped access. Never place it in React, Helm values or Git.

Keycloak brokers the BAC identity provider using the agreed OIDC or SAML connection. The corporate provider owns credentials, MFA and user lifecycle. React is not registered as the OAuth client for this BFF-based web flow. Machine callers use separate clients. One logical protected API audience, `aobp-api`, is selected initially for the governed AOBP API trust boundary; every participating service explicitly validates it. Separate sensitive domains may require separate audiences and deliberate token exchange. Merely registering audiences does not make services secure.

Step 2 - Load the separately deployed UI

The browser downloads the shell and MFE assets through ALB static routes. React calls `GET /bff/api/v1/session`. ALB sends it to gateway; gateway forwards it unchanged to BFF. With no valid session, the BFF returns a JSON 401. React displays a sign-in action. It does not receive an HTML identity login page inside a fetch response.

Step 3 - Start login with top-level navigation

The sign-in action navigates to `/oauth2/authorization/keycloak`. BFF creates a login transaction containing state, nonce, PKCE verifier, expiry and an allowlisted relative return path. Store that transaction in the shared server-side session repository before redirecting. Another BFF replica must be able to complete it. BFF redirects the browser to Keycloak, which redirects to BAC SSO if necessary. Corporate credentials are entered only at the approved identity page.

Figure 2. SSO login from first visit to authenticated session
See the landscape diagram plate following this section.

Step 4 - Complete SSO and exchange the code

After successful login, Keycloak sends the browser to the registered BFF callback with a code and state. The request again follows ALB → gateway → BFF. BFF checks the stored transaction, state and expiry, then exchanges the one-use code server-to-server using PKCE and client authentication. Validate returned ID-token signature, issuer, audience, nonce and timestamps using the OIDC library. Do not hand-roll the protocol.

The BFF receives tokens over its secured back channel, rotates the session identifier after authentication, and saves the session and authorized client. The browser receives only the session cookie and a redirect to a clean relative application URL. Disable raw access logging on the browser ALB as specified in section 17; redact callback query strings at gateway, BFF, analytics and error-reporting sources; set an appropriate Referrer-Policy and avoid third-party content on the callback response.

Login sequence: who does what

Step	Actor and action	What the reader should understand
1	Browser loads React and asks BFF for session status	Downloading the screen does not grant access to data
2	BFF returns 401 when no session exists	The screen offers Sign in; an API fetch does not become an invisible login page
3	Browser navigates to BFF login through ALB and gateway	BFF records a short-lived login transaction in shared storage
4	BFF redirects through Keycloak to BAC SSO	The password and MFA are handled by the corporate identity system
5	Keycloak returns a code to the exact BFF callback	The callback can land on a different healthy BFF replica
6	BFF validates the transaction and exchanges the code privately	Tokens are returned to the BFF over the back channel
7	BFF saves tokens and rotates the session identifier	Browser receives an opaque cookie and a clean application redirect
8	React obtains session/capabilities and a fresh CSRF token	The screen can now request authorized data; APIs still check every action

Step 5 - Protect the session and CSRF boundary

Use a host-only session cookie such as `__Host-AOBPSESSION`, `Secure; HttpOnly; Path=/`, with no Domain attribute. **SameSite=Lax is the selected compatibility choice for the shared pre-login/session cookie and top-level GET OIDC callback.** This is a documented relaxation from the stricter SameSite preference: an external IdP must return the correlation cookie to the callback. Do not use cross-site form_post with this assumption. If Strict is required, implement a separately bounded login-correlation mechanism and test the complete redirect chain. Never relax to None merely to make a broken route work.

Use the stronger `__Host-Http-` cookie prefix where the supported browser matrix enforces it; the examples retain `__Host-AOBPSESSION` for compatibility. Both require explicit Secure and HttpOnly configuration, host-only scope and Path=/. Because Path=/ also sends the cookie on same-host static requests, static workloads must ignore it, never log it and never treat it as permission to embed operational data in assets.

Set initial idle timeout to 30 minutes and absolute session lifetime to 8 hours, bounded by the IdP/refresh-token lifetime. These are AOBP proposals, not industry constants. A cookie can outlive its server record; the server always decides validity. Refresh, WebSocket heartbeats and background polling must not reset the absolute lifetime; define which genuine user activity extends idle time.

Expose `GET /bff/api/v1/csrf` as a no-store, same-origin CSRF bootstrap bound to the session. React sends the returned token in `X-CSRF-TOKEN` for POST/PATCH/PUT/DELETE, including logout. Keep CSRF protection enabled, validate exact Origin where applicable, and refresh the CSRF token after login/logout rotation. Use the supported Spring Security SPA integration for the selected version, including its deferred/masked-token behavior. No wildcard credentialed CORS; normal same-origin requests need none.

Step 6 - Process an authenticated business request

Figure 3. An authorized business request and its return path
See the landscape diagram plate following this section.

1. React sends `GET /bff/api/v1/flights?terminalId=T1`. The browser automatically includes the AOBP session cookie over HTTPS.
2. ALB selects a healthy gateway pod. Gateway validates the route and request limits and forwards to a healthy BFF pod.
3. BFF resolves the session in the shared store, checks lifetime and session revocation, and checks CSRF for unsafe methods.
4. BFF obtains this session's authorized client, checks token freshness and performs a coordinated refresh if needed.
5. BFF maps the approved browser contract to the private Flight API. It removes browser cookies and untrusted identity/authorization headers, then sets `Authorization: Bearer <access-token>` and approved tracing headers.
6. Flight API independently validates signature, algorithm, issuer, `aud=aobp-api`, expiry/not-before and required client/scope constraints. An ID token is never accepted as an API access token.
7. The service resolves current business permissions and checks the requested action plus trusted resource attributes. For a list, authorized terminal scope is applied before pagination, counts and exports.

8. Service returns only authorized data. BFF shapes the browser response, strips unsafe downstream headers and returns via gateway and ALB. The access token never appears in the JSON response.

An access token goes downstream in an **Authorization header**, not a cookie, URL or JSON field. A refresh token never goes to a business service. No service calls the browser or BFF to refresh a token.

Browser and server request example

These two requests represent the same user action at two different trust boundaries. Values below are placeholders, never real credentials.

HTTP · illustrative specification

```
GET /bff/api/v1/flights?terminalId=T1 HTTP/1.1
Host: aobp.example.com
Cookie: __Host-AOBPSESSION=<opaque-session-reference>
```

After session validation, the BFF constructs a new private request. It does not copy the browser Cookie header downstream.

HTTP · illustrative specification

```
GET /api/v1/flights?terminalId=T1 HTTP/1.1
Host: flight-api.aobp-apps.svc.cluster.local
Authorization: Bearer <server-held-access-token>
traceparent: <validated-trace-context>
```

For PATCH/POST/PUT/DELETE, React also sends its session-bound `X-CSRF-TOKEN` to the BFF. The BFF forwards only an allowlist of application headers and constructs authorization itself. All application hops use the transport controls in section 18. Redaction must apply to both HTTP examples in diagnostics.

A simple example: one click, two different credentials

An operator clicks a flight in the React screen. The browser sends its session reference to the BFF through ALB and gateway. The BFF looks up the operator's session and privately retrieves the API token. It sends that token to Flight API, which checks identity, permission and terminal scope. The response contains the permitted flight information. The browser never receives the API token, and Flight API never receives the browser's session cookie.

Event the user notices	What happens behind the screen	User-visible result
First visit	No BFF session; UI starts approved SSO navigation	Corporate sign-in, or immediate return if SSO is already active
Normal click	Session valid; access token fresh; service permits action	Requested authorized data appears
Access token near expiry	BFF refreshes under coordinated control before the API call	Usually no interruption
Session expired or refresh definitively rejected	BFF removes unusable state and returns a safe 401	UI offers sign-in again; never loops silently
Authenticated user lacks permission	API returns 403 after its own checks	Clear access-denied message; no refresh attempt

Event the user notices	What happens behind the screen	User-visible result
Identity or session dependency unavailable	Bounded failure path; no credential bypass	Clear temporary-unavailable message with support reference

Step 7 - Run distributed BFF replicas correctly

Figure 4. Shared sessions and coordinated refresh across BFF replicas
See the landscape diagram plate following this section.

Use Spring Session Data Redis for clustered HTTP sessions, including pre-login correlation. Configure an explicit OAuth2AuthorizedClientRepository/Service that persists access/refresh/ID-token state in a protected shared store associated with the session and client registration. Session storage alone does not prove authorized-client persistence. Avoid an in-memory repository and avoid sharing one user's tokens solely by username across unrelated sessions.

Session metadata includes session ID/reference, issuer+subject, createdAt, idle/absolute expiry, revocation/session generation and the authorized-client reference. Authorized-client data includes registration, token expiry, current encrypted credential values and a version. Redis-compatible storage uses TLS, private endpoints, encrypted storage, least-privilege ACLs and controlled administration. Protect credentials with application-level envelope encryption where required by the platform threat model; store wrapping keys in KMS. Verify backups, diagnostics and support tooling cannot expose raw tokens.

Any healthy BFF can serve the next REST request; no sticky session is needed. A live WebSocket still belongs to one pod. Keep long-lived socket state separate from shared HTTP session records.

Step 8 - Refresh before expiry and coordinate concurrent requests

Use OAuth2AuthorizedClientManager with authorization_code and refresh_token providers and configured success/failure persistence handlers. Before an outbound API call, treat a token inside a proposed 60-second expiry safety window as needing refresh. Use a window suitable for request duration and clock skew; a fixed minute is not a guarantee for all workloads.

1. Acquire a bounded per-session/per-client refresh lease, then reread the authorized-client record.

2. If another pod already refreshed it, use the newer version.

3. Otherwise call Keycloak's token endpoint using the refresh token and BFF client authentication. Persist rotated refresh and access tokens together with a version-checked update.

4. Use an ownership-checked lease release and prevent stale holders overwriting a newer record. Lease expiry, pod death and rotating-token races require explicit tests; a basic SETNX lock alone is insufficient.

5. Bound waiting callers. A definitive invalid_grant/revocation invalidates the session. A temporary IdP outage produces controlled 503 when no usable token remains, rather than falsely treating the user as revoked.

6. A timeout after a refresh may leave its result unknown. Do not repeatedly replay an old rotating refresh token. Follow the selected IdP's documented reuse behavior; if safe reconciliation is impossible, require a fresh login.

Refresh sequence: a plain-language example

Assume the person is still within the permitted session lifetime, but their short-lived API access pass is about to expire. The person does not need to type their password for every renewal. The BFF renews the pass privately, while keeping the browser cookie unchanged unless session policy requires rotation.

Step	BFF and shared-store action	Failure behavior
1	Load this session's current token version and expiry	Missing or expired session returns 401; store outage returns controlled 503
2	If refresh is needed, coordinate one owner and reread the record	Waiting requests have a bounded deadline
3	Owner asks Keycloak for new tokens using the protected refresh token	Definitive revocation requires login; transient outage does not imply revocation
4	Atomically save new access and rotated refresh tokens with a version check	A stale worker cannot overwrite newer state
5	All waiting callers use the committed new version	No refresh token is sent to a microservice or React

Redis replication failover can lose a recent lease or token update. Version checks protect local writes but cannot cancel a token request already accepted by Keycloak. Therefore a Redis lease is not a promise of exactly-once refresh under every failure. Bound retries, test failover with refresh rotation, and require a new login when the credential outcome cannot be reconciled safely. A stricter exactly-once requirement needs an identity-compatible durable coordination design, not a longer lock timeout.

Step 9 - Handle expiry during processing

JWT validation normally happens when each request arrives. If the token expires after a service has accepted a request, it does not automatically interrupt the running transaction. Long-running workflows use explicitly authorized durable jobs rather than keeping the user's token alive indefinitely. High-risk operations may recheck current permission/revocation immediately before commit.

If Flight API then calls Workflow API, Workflow API performs its own entry validation. A token accepted by the first service might now be expired at the second. Keep synchronous chains short and refresh with an adequate margin at BFF. Define client-credential identities for background jobs, or explicit token exchange for narrower delegated audiences; do not silently elevate a failed user request into a machine request.

A downstream 401 does not always mean expiry: issuer, audience or signature may be wrong. Never refresh on 403. For a confirmed authentication expiry, allow at most one refresh/retry for a safe read, or for a mutation protected by a durable idempotency contract. Do not blindly replay a command or distributed chain whose outcome may already be committed. Return an intelligible error and let the client reconcile operation state where the outcome is unknown.

Step 10 - Logout, disablement and revocation

React submits CSRF-protected `POST /bff/logout`. BFF deletes the session and authorized-client state, clears the cookie, publishes a cross-replica socket-close/revocation notification and invokes supported IdP revocation/logout. Navigate the browser through approved end-session redirects if corporate SSO logout is intended. Define 'log out of AOBP' separately from 'log out of all BAC applications'.

Expose a dedicated POST receiver at `/bff/oidc/backchannel-logout`, registered exactly with the IdP. Gateway permits only that method and bounded form size; restrict network sources where feasible. This protocol endpoint is exempt only from browser CSRF/session requirements because it authenticates the signed logout token. It is not a general CSRF exclusion for `/bff/**`. Validate signature, issuer, audience, issued-at, required logout event, and sid or subject; reject a nonce claim and deduplicate replay using the logout token jti for its validity window. Persist the invalidation before acknowledging successful processing, and test IdP retry behavior.

Implement Keycloak back-channel logout with logout-token validation and a mapping from IdP session identifier to all affected BFF sessions. Subject-wide invalidation needs issuer+subject indexing. Avoid logging the index's raw identity data. Already issued self-contained JWTs may remain valid until expiry unless services perform current revocation/entitlement checks. A BFF logout is not instantaneous global JWT invalidation.

Propose five-minute access tokens and a maximum 60-second account-disable decision propagation target for protected actions, subject to security/business approval and test evidence. The shorter propagation target requires service-side current entitlement/revocation checks with bounded caches; JWT expiry alone cannot meet it. Privileged mutations fail closed when required fresh authorization data is unavailable. Rotating refresh tokens, socket closure and session TTL form part of the same tested lifecycle.

Logout and refresh must not resurrect a session

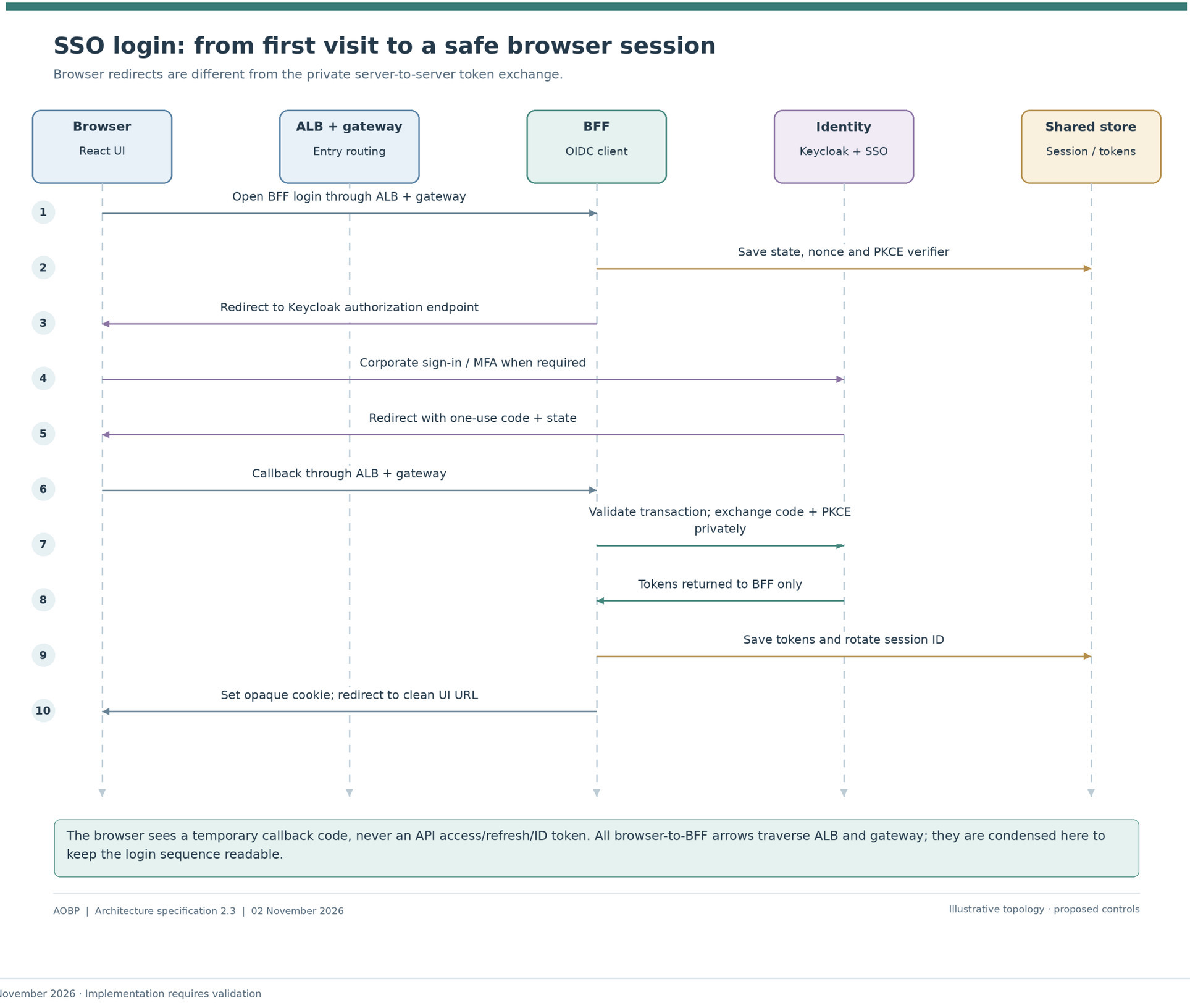
The token save operation must check that the session still exists, has the same generation, is not revoked and has not exceeded its absolute expiry. A refresh started before logout must not recreate a deleted authorized-client record afterward. Use an atomic compare-and-save against the session generation; retain a bounded revocation marker where necessary for in-flight operations. Invalidate shared state before reporting local logout success, clear the cookie, then coordinate socket closure and IdP operations. A failed remote logout must not recreate a locally invalidated session.

Step 11 - Preserve the no-local-user-accounts principle

AOBP stores roles, permissions, issuer-qualified external-group mappings, scope policies and audit references. It does not store corporate passwords or maintain a parallel editable employee directory. A BFF session is transient security state, not an application user account. Keycloak may retain federation metadata; this must be recognized explicitly in BAC's identity policy.

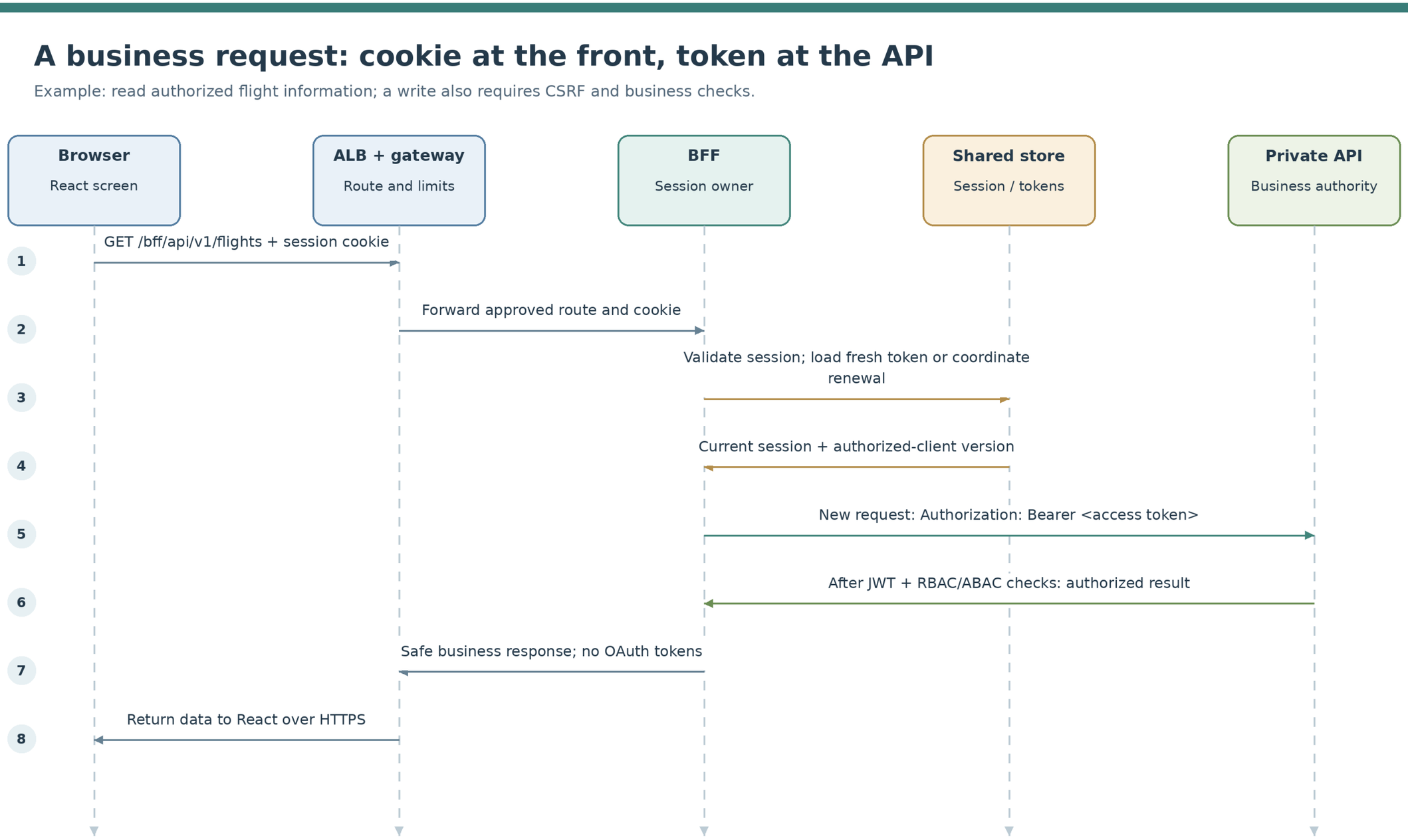
Use immutable issuer+subject and stable group identifiers. Resolve group overage, missing claims and lifecycle events through approved IdP mechanisms; absence of group data must not grant broader access. The Central Security Service maps trusted claims to AOBP permissions and terminal scopes. UI capability information is for display and never substitutes for backend checks.

Figure 2. SSO login from first visit to authenticated session



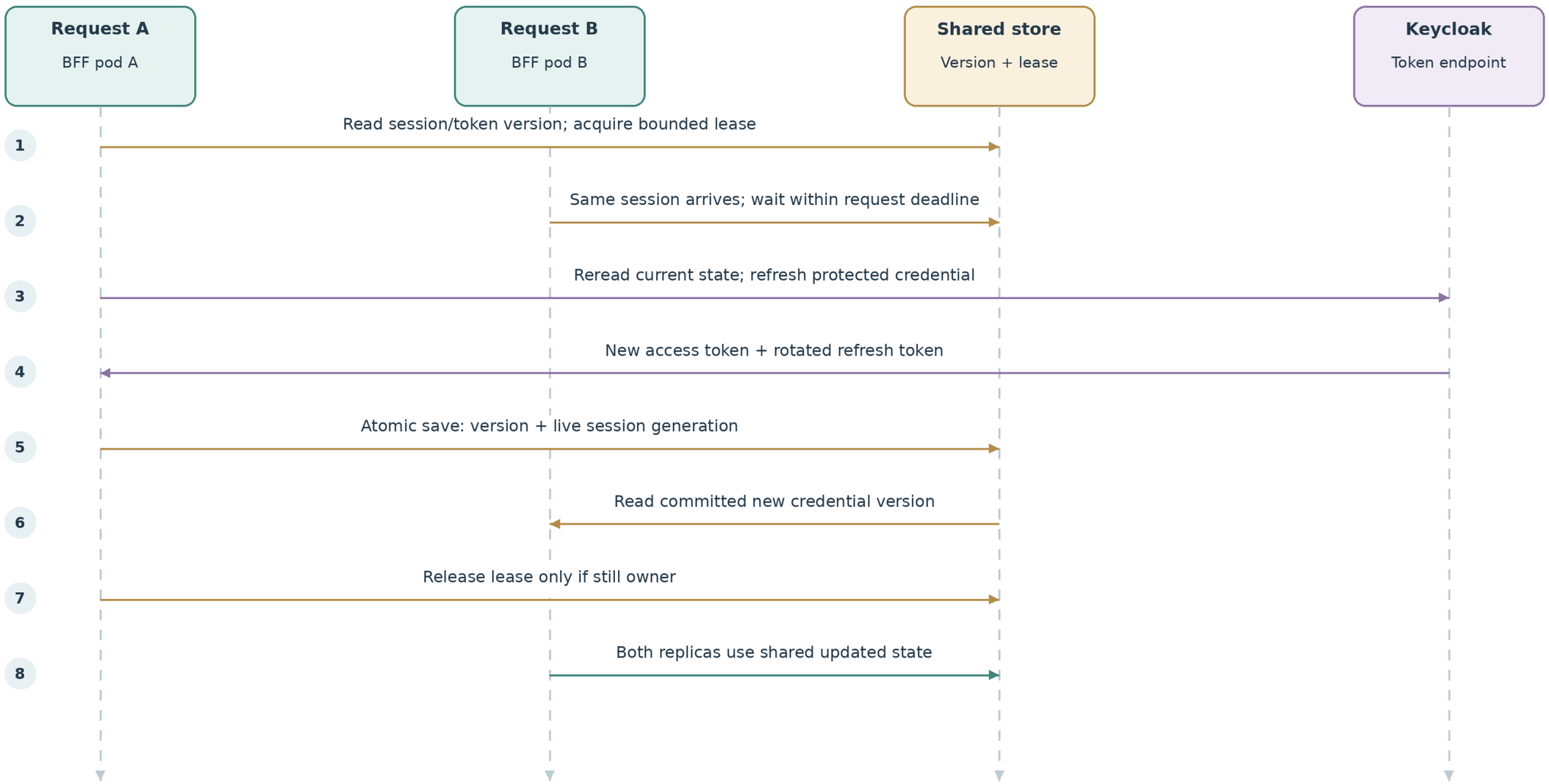
The browser sees a temporary callback code, never an API access/refresh/ID token. All browser-to-BFF arrows traverse ALB and gateway; they are condensed here to keep the login sequence readable.

Figure 3. An authorized business request and its return path



The BFF removes browser cookies and untrusted identity headers before calling an allowlisted private API. A hidden UI button is never permission enforcement. Denials and errors return through the same route.

Figure 4. Shared sessions and coordinated refresh across BFF replicas



7. Business authorization

Layer	Checks
Edge	Network/request protections and coarse bearer scope for partner routes
BFF	Authenticated session, CSRF, approved routes and subscription controls
Flight API	Permission, object scope, field access and business invariant
Query layer	Authorized predicate before pagination/count/export
Workflow API	Current task eligibility, resource scope, claim/complete/reassign permissions

Example permissions: `flight.read`, `flight.operational-time.update`, `flight.export`, `task.claim`, `task.complete`. Designer/publisher permissions are independent from operational privileges. A studio manifest declares required permissions but cannot grant them to users.

ABAC input: validated subject, current entitlements, resource airport/terminal/airline from the database, requested action, session assurance where needed, and current flight/task state. Flight terminal and identity headers from the browser are never authoritative. Effective list scope is authorized scope intersected with requested filters. Deny by default.

Reference interface: `FlightAuthorization.checkCanRead(subject, flight)` and `checkCanUpdate(subject, flight, requestedChanges)`. Implement initially with tested Spring policy classes or an OPA adapter. For polyglot reuse, distribute versioned policies to local evaluators. Supply dynamic resource data from the owning service. Define freshness bounds; deny protected actions if necessary policy/entitlement data is unavailable or stale.

Recheck update invariants under transaction/optimistic locking. A permission allow does not override an invalid business state. Use idempotency for retried commands. Do not automatically retry non-idempotent mutations at the gateway.

Central Security Service and a simple worked example

The Central Security Service is the shared policy decision capability; services remain enforcement points. It owns authorization configuration, not corporate authentication. Gateway may use cached coarse operation mappings for partner routes. Browser BFF routes rely on BFF session checks; the private API always performs the authoritative business check. An authenticated private service call must not be treated as authorized just because it originated in the cluster.

Example: User A belongs to BAC Operations, maps to OPERATOR, and may work on Terminal T1. Updating a T1 flight requires both flight.operational-time.update and a matching terminal scope. Updating a T2 flight returns 403 even when the request passed ALB, gateway and BFF. Changing terminalId in the browser request cannot change the trusted terminal read from the flight database.

Permission mapping uses business capabilities rather than one permissions table per microservice. A logical API audience is still registered with the IdP. Absence of an application-owned microservice registration table does not exempt APIs from client registration, audience checks or workload identity.

Central policy availability is an operational concern. Use versioned policy/config caches or local evaluators where appropriate; define freshness explicitly. Do not silently allow writes when the security

service is unavailable. Record deny decisions and policy version without copying tokens or full identity profiles into logs.

8. WebSocket flow and scaling

Browser connects to `wss://aobp.example.com/bff/ws/v1` after login. The gateway proxies the upgrade; BFF authenticates the HTTP principal, validates Origin, and associates the connection with the session. With Spring STOMP security, send CSRF in CONNECT and authorize destinations/messages. Native WebSocket requires equivalent application controls.

On `SUBSCRIBE`, authorize the topic/resource against current scope. On event delivery, ensure recipients remain entitled, including role changes and changes to a flight's terminal/airline. Never broadcast all airport events and rely on React to hide them. Client-supplied destinations are not authorization evidence.

Flight API commits update + outbox record. Relay publishes a versioned event with event_id/business key. BFF receives it, finds authorized local subscriptions, and sends a minimal permitted payload. Use event ID and resource version to deduplicate and detect out-of-order changes. For complex payload security, send a scoped invalidation and let the browser re-fetch an authorized snapshot.

Multiple BFF replicas require fan-out to every replica holding relevant connections, or a broker relay/subscription architecture that accomplishes it. One Kafka consumer group delivers an event to one consumer, not every BFF. Do not use a single shared group as a broadcast design. Shared Redis sessions do not share live sockets or subscriptions.

Connections remain attached to their accepting instance. Native WebSocket generally needs no stickiness after upgrade; fallback transports may. Use graceful draining, connection/message limits, heartbeats, bounded buffers and slow-consumer disconnects. Reconnect with exponential backoff/jitter and obtain a fresh snapshot or replay cursor. Sockets are not a durable record of business state.

Map session expiry/logout to local socket closure across replicas; define bounded periodic checks because session notifications can be delayed. Server push/heartbeats must not keep a user logged in indefinitely. Define idle and absolute lifetime independently.

9. API naming and resource modelling

The rules in this chapter are AOBP platform conventions. They adopt HTTP, JSON, OpenAPI and Problem Details standards, but conventions such as URL versioning, camelCase fields and success envelopes are local choices, not universal industry mandates. MUST indicates an acceptance requirement; SHOULD allows a documented exception approved by the API owner and security/platform reviewer.

URI standard

Boundary	Canonical format	Example
Browser BFF API	`/bff/api/v{major}/{resources}`	`/bff/api/v1/flights`
Domain API, private or partner	`/api/v{major}/{resources}`	`/api/v1/flights`
Browser native WebSocket endpoint	`/bff/ws/v{major}`	`/bff/ws/v1`
BFF framework login endpoints	Framework-defined stable routes	`/oauth2/authorization/keycloak`
BFF framework callback	Exact registered redirect URI	`/login/oauth2/code/keycloak`
Health and telemetry	Private management endpoints	`/actuator/health/readiness`, `/actuator/prometheus`
Versioned MFE assets	`/mfes/{name}/{release}/...`	`/mfes/flight/1.4.0/remoteEntry.js`

Hostnames differ by environment; path contracts do not. Use `https://aobp.example.com` as the illustrative browser origin and `https://identity.example.com` as the issuer host. Partner APIs may use `https://api.example.com` while preserving `/api/v1/...`. Internal DNS names and ports never appear in browser contracts.

Paths MUST use lowercase plural resource nouns and kebab-case for compound segments. JSON/query/property names MUST use lowerCamelCase. OpenAPI path parameters use `{flightId}`; runtime requests contain an actual identifier, not braces. Avoid trailing slashes and ambiguous alternate spellings. IDs are opaque, immutable, non-PII references: a flight number plus date is not a reliable unique flight identifier. UUID is the reference representation; choose the UUID generation variant during implementation.

Do not include a service framework, deployment location, user identity, password, bearer token or action verb such as `getFlights` in the URI. Scope is enforced from trusted identity and resource data; a path airportId is not evidence of access. Keep nesting shallow and use nested paths only where ownership is real. URI versions identify breaking API contracts, not application releases. Additive compatible changes stay in v1; breaking changes need a new major contract and coexistence/migration plan.

Resource catalogue

Method and path	Meaning	Success
`GET /api/v1/flights`	Authorized filtered flight collection	200
`GET /api/v1/flights/{flightId}`	One authorized flight representation	200 + ETag
`GET /api/v1/flights/{flightId}/operational-times`	Read the operational-time subresource and concurrency token	200 + ETag
`PATCH /api/v1/flights/{flightId}/operational-times`	Partial update of defined operational-time fields	200 + new ETag
`POST /api/v1/flight-change-requests`	Create a governed operational change request	201 + Location
`POST /api/v1/flight-change-requests/{requestId}/approvals`	Record an authorized approval	201 + Location
`POST /api/v1/flight-exports`	Request an asynchronous export	202 + Location
`GET /api/v1/operations/{operationId}`	Read asynchronous operation status	200
`GET /api/v1/flights/{flightId}/events`	Authorized durable business history, if exposed	200

These examples define the reference Flight API surface; each operation is implemented only if the business capability exists. POST does not imply blanket approval to create flight records. GET/HEAD MUST be safe; never use GET to update state or log out. PUT replaces the documented representation; PATCH updates explicitly permitted fields; DELETE means the documented deletion semantics, not a hidden arbitrary command. Use subordinate command resources for approvals/exports with lifecycle and audit needs.

BFF routes mirror or compose domain operations. One example mapping is `/bff/api/v1/flights/{flightId}` to `/api/v1/flights/{flightId}`. Do not blindly forward all HTTP paths/headers. Composition endpoints such as `/bff/api/v1/flight-dashboard` may combine responses while retaining explicit schemas and partial-failure semantics.

Filtering, pagination and sorting

Example: `GET /api/v1/flights?airportId=BRU&terminalId=T1&airlineCode=SN&scheduledFrom=2026-10-30T00%3A00%3A00Z&scheduledTo=2026-10-31T00%3A00%3A00Z&limit=50&sort=scheduledTime,flightId`.

Use RFC 3339 timestamps and a half-open time interval `[scheduledFrom, scheduledTo)`. `airportId=BRU` is an illustrative configured airport key; distinguish it from `airportIataCode` or `airportIcaoCode` if those attributes are exposed. Validate named filters; never accept raw SQL, column names or arbitrary expressions. Sort keys are allowlisted; prefix descending keys with `-`. Always append a unique tie-breaker.

Cursor pagination is the reference for potentially large flight collections. `limit` defaults to 50 and is bounded at 200 as an initial platform convention; confirm against payload sizes and load. Cursor is opaque, integrity-protected, expires under a documented policy, and binds filters, sort and applicable authorization context. Reauthorize every page. Do not silently upgrade requested scope. Reject incompatible/tampered cursors with 400. Avoid mandatory total counts on large queries; use an optional separately budgeted count when needed.

Changing flight schedules can move records between pages. A normal cursor over live data is not a snapshot guarantee. For consistent exports, use a server-side snapshot/as-of strategy and a job. Restrict query ranges, export volume and execution cost.

Third-party REST integration: step by step

Figure 6. Dedicated machine identity and gateway-to-API routing
See the landscape diagram plate following this section.

The partner is another computer system. It identifies itself with its own approved credential; it does not borrow an airport employee's browser session.

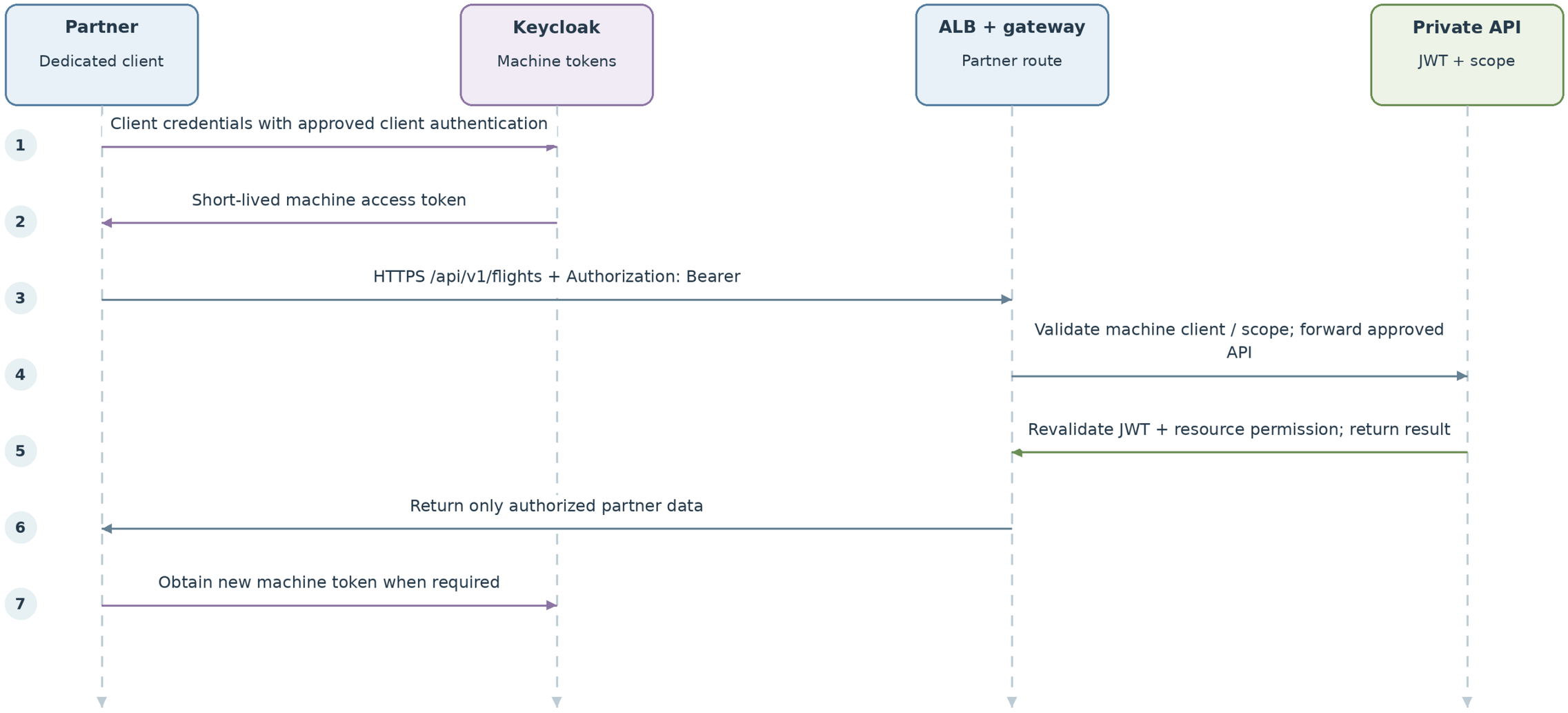
1. Register a dedicated machine client for the partner in the IdP; grant only required API scopes. Use asymmetric client authentication where supported and rotate credentials. Do not distribute a human refresh token or BFF cookie.
2. The partner authenticates to the token endpoint using client_credentials and requests the agreed API resource/scope. The resulting principal is the system, not an impersonated airport operator.
3. It calls `https://api.example.com/api/v1/flights` with Authorization: Bearer. ALB forwards to the partner gateway route; gateway validates the token and coarse scope.
4. Flight API validates the token again, identifies the approved client and enforces its airport/terminal/data scope and permitted operations. Machine identities must not inherit ADMIN merely because there is no human group claim.
5. The response returns through the gateway and ALB. The partner obtains a new machine token when needed; the BFF is not involved.

For outbound AOBP-to-partner REST, the owning integration service uses its own approved machine credential from the secret manager. Do not forward AOBP user tokens to arbitrary partner endpoints. Delegated user integration requires explicit federation/token exchange and a separate contract. Webhooks use a dedicated authenticated receiver and replay controls, not a cookie-session CSRF exception across all browser routes.

Figure 6. Dedicated machine identity and gateway-to-API routing

Machine integration: its own identity and API route

A partner system does not sign in as an airport employee and does not use the browser BFF.



No BFF session, employee password or human refresh token is shared. Partner routes require approved clients and scopes; a different hostname or CORS configuration alone is not an authentication control.

10. Requests, responses and headers

Representation conventions

Use UTF-8 JSON with `application/json` for success responses; JSON Merge Patch uses `application/merge-patch+json`. Use a documented data envelope for business JSON responses. Errors MUST use the top-level Problem Details object and MUST NOT be wrapped in `data` or a second error envelope. Health probes, OIDC protocol responses, metrics, files, streams and 204/304 responses retain their protocol-defined representations.

Use lowerCamelCase fields, JSON booleans, numbers with documented bounds/precision, ISO dates and UTC RFC 3339 timestamps ending Z. Represent exact decimal monetary/measurement values as documented decimal strings when binary floating point would be unsafe; include units/currency. Enum values use UPPER_SNAKE_CASE and are explicitly documented. Define nullability per field: null means explicitly unknown/not available where allowed; absent means not included or not applicable according to that schema. Empty collections are `[]`, not null. Do not serialize ORM entities or expose internal columns.

Reference single-flight response:

JSON · illustrative specification

```
{
  "data": {
    "flightId": "f47ac10b-58cc-4372-a567-0e02b2c3d479",
    "flightNumber": "SN1234",
    "airportId": "BRU",
    "terminalId": "T1",
    "movementType": "DEPARTURE",
    "operationalStatus": "SCHEDULED",
    "scheduledTime": "2026-10-30T16:30:00Z",
    "estimatedTime": "2026-10-30T16:40:00Z",
    "actualTime": null,
    "version": 17
  }
}
```

Reference collection response:

JSON · illustrative specification

```
{
  "data": [
    {
      "flightId": "f47ac10b-58cc-4372-a567-0e02b2c3d479",
      "flightNumber": "SN1234",
      "scheduledTime": "2026-10-30T16:30:00Z"
    }
  ],
  "meta": {
    "limit": 50,
    "hasMore": true,
    "nextCursor": "opaque-integrity-protected-cursor"
  },
  "links": {
    "next": "/api/v1/flights?limit=50&cursor=opaque-integrity-protected-cursor"
  }
}
```

Collection items use an explicit summary schema, not an accidentally truncated detail schema. BFF emits BFF-facing links and its own documented DTOs; do not leak internal URLs. At the last page use hasMore=false, nextCursor=null and omit links.next. Do not repeat HTTP status, success=true or

generic messages in every successful body. Use headers for request correlation; do not add timestamps and trace metadata to all business payloads unnecessarily.

Headers

Header	Rule
`Content-Type`, `Accept`	Validate supported media types; 415/406 where appropriate
`Authorization: Bearer ...`	BFF-to-API and partner API only; redact everywhere
`Cookie` / `Set-Cookie`	BFF browser session only; redact everywhere
`traceparent`, `tracestate`	W3C trace propagation on allowlisted boundaries
`X-Request-ID`	Sanitized opaque HTTP interaction identifier returned to caller
`Idempotency-Key`	AOBP application contract for specified POST operations, not a guarantee supplied by HTTP itself
`ETag` / `If-Match`	Optimistic concurrency for versioned resources
`Location`	Newly created resource or asynchronous status URL
`Retry-After`	Applicable rate-limit/unavailable responses; honour before retry
`Cache-Control`	Sensitive/session responses default no-store; caching must be explicitly designed
`WWW-Authenticate`	Preserve appropriate bearer challenges on API 401

Generate/validate X-Request-ID at the trusted edge; internal spans may carry that value for a synchronous chain. A trace identifies distributed execution and may include retries; request ID identifies the interaction. Do not use either as user identity or an idempotency key. Do not expose arbitrary baggage fields. The BFF may return a correlation header but MUST NOT forward Set-Cookie/Authorization from unrelated downstream services.

Concurrency and PATCH

For the reference operational-times subresource, GET the subresource and receive a strong opaque ETag. PATCH requires If-Match and the documented merge-patch schema. Missing required precondition returns 428; stale If-Match returns 412. Return the new representation and new ETag on success. Never synthesize an ETag solely from a guessed client version. Authorize before exposing resource/version information.

The patch DTO allowlists fields such as estimatedTime. Explicitly reject forbidden and unknown input fields to prevent mass assignment. Define whether null clears a nullable field; reject null for fields that cannot be cleared. Use a transaction and version check so the conditional write and state invariants are atomic. Return 409 for genuine business-state conflicts, distinct from a failed HTTP precondition. GET flight detail may use its own representation ETag; do not assume it is interchangeable with a subresource ETag.

Asynchronous operations and idempotency

POST export returns 202 and Location `/api/v1/operations/{operationId}`. Status resource uses QUEUED/RUNNING/SUCCEEDED/FAILED/CANCELLED with createdAt/updatedAt and an authorized result reference when ready. A failed job's status GET can still return 200 because retrieving status succeeded; its documented error field describes the job failure. Result download rechecks authorization, expiry and data scope; private signed URLs are short-lived and never logged.

For selected POSTs, require an Idempotency-Key. Scope persistence by caller identity/organization, operation and key; store a canonical request hash and result in durable storage with a uniqueness constraint. Concurrent duplicate execution must be fenced. Same key/same request replays the documented original outcome; same key/different request returns 409. In-progress duplicates return the existing operation reference or a documented conflict. Start with a 24-hour retention proposal and agree it against client retry horizons. Do not claim idempotency after expiry. Never use pod memory as the only store. Reauthorize a replay and avoid storing secrets or unrestricted response bodies in the idempotency record.

11. Exceptions and error responses

Adopt RFC 9457 Problem Details for HTTP APIs, with Content-Type `application/problem+json`. Standard members are type, title, status, detail and instance. AOBP extensions are code, requestId, traceId and optional errors. Problem type URLs identify stable documentation, not internal stack traces. They should resolve to controlled API documentation. The public hostname below is illustrative.

JSON · illustrative specification

```
{
  "type": "https://api.example.com/problems/validation-failed",
  "title": "Request validation failed",
  "status": 400,
  "detail": "One or more request fields are invalid.",
  "instance": "urn:uuid:8e6699b3-5c73-4c72-9f56-1c68568b0918",
  "code": "AOBP.VALIDATION_FAILED",
  "requestId": "req_01J_EXAMPLE",
  "traceId": "4bf92f3577b34da6a3ce929d0e0e4736",
  "errors": [
    {
      "location": "body",
      "pointer": "/estimatedTime",
      "code": "INVALID_TIMESTAMP",
      "message": "Use a valid RFC 3339 timestamp."
    }
  ]
}
```

For query/path/header errors, use location plus parameter name instead of pretending there is a JSON Pointer. Do not include rejected secret/PII values. instance is a unique error occurrence identifier; it need not be a fetchable URL. traceId is present only when known and safe to disclose. Clients branch on code/type, never localized title/detail. Keep titles stable for a problem type.

HTTP	Code example	Trigger / handling
400	AOBP.VALIDATION_FAILED	Malformed JSON, invalid fields, filters or cursor
401	AOBP.AUTHENTICATION_REQUIRED	Missing/expired session or invalid bearer token
403	AOBP.ACCESS_DENIED	Authenticated but lacks action/scope
404	AOBP.RESOURCE_NOT_FOUND	Absent resource; also concealed inaccessible resource by agreed policy
405	AOBP.METHOD_NOT_ALLOWED	Include Allow header
409	FLIGHT.STATE_CONFLICT	Current business state disallows action
412	AOBP.PRECONDITION_FAILED	Stale If-Match
413	AOBP.PAYLOAD_TOO_LARGE	Request exceeds documented size
415	AOBP.UNSUPPORTED_MEDIA_TYPE	Unsupported request representation
428	AOBP.PRECONDITION_REQUIRED	Required If-Match absent
429	AOBP.RATE_LIMIT_EXCEEDED	Include Retry-After where known
500	AOBP.INTERNAL_ERROR	Unexpected application defect; safe generic detail

HTTP	Code example	Trigger / handling
502	AOBP.UPSTREAM_RESPONSE_INVALID	Gateway/BFF receives invalid upstream response
503	AOBP.SERVICE_UNAVAILABLE	Necessary dependency unavailable/circuit open
504	AOBP.UPSTREAM_TIMEOUT	Proxy deadline exceeded waiting for upstream

These are platform mappings, not proof all failures have one universal status. A domain invariant outside the current state may use a separately defined 422 contract; do not use 400/409/422 inconsistently for the same condition. Preserve actual HTTP status; never return 200 with success=false for an HTTP operation failure.

Spring implementation pattern

Component	Implementation responsibility
Controller	Validated request DTOs, ResponseEntity, documented headers and status
Application service	Use-case orchestration and transaction boundary
Domain	Typed business exceptions with stable codes, not raw HTTP coupling
Policy adapter	Typed authorization decision; trusted attributes
REST exception handler	`@RestControllerAdvice`, ResponseEntityExceptionHandler, Spring ProblemDetail
Security entry/deny handlers	Same JSON contract for filter-chain 401/403
Gateway handler	Reactive error handler for routing/timeouts; separate security handlers
BFF downstream mapper	Preserve approved domain errors; transform unsafe/unknown payloads
WebSocket layer	Message-level error contract after upgrade

MVC advice alone does not handle gateway reactive errors, security-filter failures or post-upgrade WebSocket errors. Implement those boundaries explicitly. Unexpected exceptions are logged once at the owning error boundary with the stack trace and returned as sanitized 500. Upstream layers record the dependency failure without repeating the full stack trace. Expected validation/business denials do not require ERROR stack traces. Do not catch Exception and return null or success.

ProblemDetail example for a business conflict (illustrative Spring code):

JAVA · illustrative specification

```
@ExceptionHandler(FlightStateConflictException.class)
ResponseEntity<ProblemDetail> handleConflict(
        FlightStateConflictException ex) {
    ProblemDetail p = ProblemDetail.forStatusAndDetail(
        HttpStatus.CONFLICT,
        "The flight cannot be changed in its current state.");
    p.setType(URI.create(
        "https://api.example.com/problems/flight-state-conflict"));
    p.setTitle("Flight state conflict");
    p.setProperty("code", "FLIGHT.STATE_CONFLICT");
```

```
    // Common factory adds instance, requestId and current traceId.
    return ResponseEntity.status(HttpStatus.CONFLICT).body(p);
}
```

Use a shared ProblemFactory and catalogue, with independent implementations for MVC and reactive stacks. Do not pass ex.getMessage() through to the caller without an explicit safe-message contract. Make framework-generated body parse, validation, unknown route, 401/403 and method errors conform too. A proxy/load balancer may still emit a generic transport error; frontend clients must tolerate non-JSON failures safely rather than crashing during JSON parsing.

12. Structured logging and audit

Canonical JSON log schema

Emit one JSON object per line to stdout/stderr, not multi-line text. Stack traces are escaped strings inside the JSON object. The following schema is an AOBP JSON export representation; OTel collectors map it to the OTel LogRecord's Timestamp, SeverityText, Body, TraceId, SpanId, Resource and Attributes. Dotted keys below are literal flat JSON keys. Do not mix a flat field and nested object with the same name in the backend index.

JSON · illustrative specification

```
{
  "timestamp": "2026-10-30T16:30:12.345Z",
  "severity": "INFO",
  "message": "Flight operational times updated",
  "event.name": "flight.operational_times.updated",
  "service.name": "flight-api",
  "service.version": "1.4.0",
  "deployment.environment.name": "production",
  "site.id": "onprem-primary",
  "trace_id": "4bf92f3577b34da6a3ce929d0e0e4736",
  "span_id": "00f067aa0ba902b7",
  "request.id": "req_01J_EXAMPLE",
  "http.request.method": "PATCH",
  "http.route": "/api/v1/flights/{flightId}/operational-times",
  "http.response.status_code": 200,
  "duration_ms": 43,
  "business.action": "flight.operational-time.update",
  "business.outcome": "SUCCESS",
  "resource.type": "flight",
  "resource.id": "f47ac10b-58cc-4372-a567-0e02b2c3d479",
  "airport.id": "BRU",
  "terminal.id": "T1",
  "policy.version": "flight-policy-7"
}
```

Always include timestamp, severity, message, event.name, service.name/version and environment. Include trace_id/span_id only when an active trace exists; never invent placeholder IDs. Include workflow.instance.id, task.id, event.id and causation.id when relevant. Actor reference is pseudonymous and restricted to justified events; never log email/name routinely. Business identifiers and terminal details also need access/retention review.

Use an allowlist for logged HTTP headers; exclude Authorization, Cookie, Set-Cookie, CSRF tokens, credentials and signed URLs. Do not log complete query strings, SQL parameters, raw request bodies, process variables or upstream payloads by default. Normalize untrusted strings, enforce length limits and prevent log injection. Schema validation should detect accidental field-type changes. Redact at source, then enforce collector/index controls.

Severity	Use
DEBUG	Controlled diagnostics, off by default in production
INFO	Meaningful lifecycle/completion events; normal request summaries at selected boundaries
WARN	Recoverable degradation, exhausted client quota or notable expected failure by policy
ERROR	Unexpected defect or failed required operation needing investigation

Avoid logging every method entry/exit and every normal authorization permit at INFO. Expected 4xx events are tracked through structured access/security metrics without treating them as application defects. A denied privileged action is always security-audited; it need not contain a stack trace. Add error.code, error.type and exception.stacktrace for unexpected faults only at the owner. Sampling diagnostics does not authorize sampling required audit events.

Audit versus technical logs

Audit event fields: auditEventId, occurredAt, actor subject+issuer reference, executing service, action, resource reference, authorized airport/terminal scope, outcome/reason, policy version, application release, requestId/traceId and redacted change summary. Keep a named schema and version. Separate business effective time from audit recording time. Capture failed attempts separately from transactionally committed outbox records.

Audit writer privileges are append-oriented; readers are separately authorized. Protect retention, archival, integrity and privileged administration. Define legal/data-owner retention periods rather than inventing one retention duration for all datasets. Diagnostic log backend availability must not decide whether a transaction's required audit record is durable.

Logging implementation

Provide language-specific adapters to the common schema: Logback JSON encoder/structured logging for supported Spring versions; standard JSON log adapters for Node/Python/Quarkus. Common standards do not require every service to depend on a Java library. Propagate correlation through instrumentation, MDC/context and asynchronous/reactive context; clear context after request completion to prevent cross-user contamination. Test Reactor and executor propagation explicitly; a thread-local-only approach is insufficient.

13. Reliability, security and event formats

Timeout and retry budget

Reference starting budget for a normal interactive read: API response objective below 2 seconds under agreed load; API dependency budget below 2 seconds; BFF overall deadline 3 seconds; gateway deadline 4 seconds; browser deadline 6 seconds. These are initial proposals to validate, not established SLOs. Set specific connect/read/pool-acquisition deadlines inside that budget, reserve overhead and use separate budgets for exports/long-running actions. Never use a retry that exceeds the caller's remaining deadline.

Retry only documented transient failures for safe/idempotent operations, with bounded attempts, exponential backoff and jitter. Respect Retry-After. Do not retry validation, authorization, state conflict or stale preconditions automatically. Choose one retry owner to prevent gateway x BFF x API multiplicative attempts. No transparent replay of partial WebSocket messages. Use bulkheads and bounded pools/queues; a circuit breaker is for dependency failures, not ordinary business rejection.

Request security controls

Validate body/header sizes, JSON nesting, collection limits, date ranges and upload/download limits. Use parameterized database queries; allowlist sort and patch fields. Never deserialize arbitrary polymorphic types from untrusted JSON. Restrict outbound hosts to prevent SSRF. Use exact CORS/origin policies only where necessary; common browser origin simplifies but does not replace CSRF. Return no-store on session/operational responses by default, and avoid user-specific shared-cache leakage. Reauthorize exports and event subscriptions. Threat-model XSS in shared-privilege MFEs, CSRF, object-level authorization, replay and injection.

Attack points, risks and mitigation responsibilities

Figure 7. Attack points and the controls at each boundary
See the landscape diagram plate following this section.

Attack point / threat	Possible consequence	Required control	Owner and acceptance evidence
A. Browser or compromised MFE / XSS	Malicious code acts through the logged-in session	No OAuth tokens in JavaScript; CSP, safe output handling, controlled remote releases and dependency checks	UI/security: inject test payloads; inspect browser storage and network responses
B. Forged browser request / CSRF	Unwanted change or logout	Session-bound CSRF token, exact Origin checks, safe GET semantics and secure cookie attributes	BFF: cross-site mutation fails before API execution
C. Edge or callback / spoofing, code leakage	Stolen login transaction, poisoned redirect or entry overload	Fixed callback, state/nonce/PKCE, trusted forwarding, WAF/limits, query redaction and controlled logging	Edge/IAM: forged callback/Host denied; retained records contain no codes or credentials

Attack point / threat	Possible consequence	Required control	Owner and acceptance evidence
D. BFF or token store / theft, SSRF, replay	Attacker obtains server-held credentials or calls arbitrary destinations	Private store, encryption/ACLs, workload identity, outbound host/path allowlists, rotation and bounded token lifetime	Platform/BFF: no arbitrary URL proxy; restricted Redis access and restore/failover tests
E. API / object or field authorization failure	Cross-terminal access or unauthorized change	JWT validation, current RBAC/ABAC, trusted resource attributes and DTO field allowlists	API owner: wrong audience, altered ID, terminal and forbidden field tests fail
F. Data / injection or audit tampering	Bulk disclosure, corruption or concealed changes	Parameterized SQL, query/export bounds, least-privilege DB roles and durable protected audit	Data/API: injection tests, export scope tests and audit integrity evidence
G. Live socket / stale entitlement	Events continue after logout or permission removal	Origin/handshake checks, per-subscription and delivery authorization, revocation fan-out and bounded rechecks	BFF/events: open sockets close and no unauthorized event is delivered within agreed bounds
H. Partner route / stolen machine credential	Automated access exceeds intended scope	Dedicated client/audience/scope, rotation, quotas and service resource authorization; mTLS if agreed	Integration/security: human web token and unauthorized partner scope are rejected

The BFF addresses browser exposure of OAuth tokens. It does not cure compromised browser code, weak business authorization or a compromised server. The table assigns the remaining defenses to the component that can enforce them. Apply STRIDE during implementation review: spoofing, tampering, repudiation, information disclosure, denial of service and elevation of privilege; retain testable threats and decisions rather than treating a product list as a threat model.

Event and WebSocket contracts

Use CloudEvents-style metadata when adopting CloudEvents, with specversion, id, source, type, subject, time, datacontenttype and data; do not call a custom envelope CloudEvents-compliant without validating the chosen transport binding. Reference event type `com.aobp.flight.operational-times.updated.v1`. Include resourceVersion and approved airport scope in data, plus correlation/causation metadata as defined extensions. Schemas are versioned and registered; consumers tolerate compatible additions and unknown optional attributes.

WebSocket message envelopes define protocolVersion, messageId, type, occurredAt and data. A message-level rejection contains code, safe message, relatedMessageId and request/trace reference where available; it is not an HTTP 400 after the connection has upgraded. STOMP ERROR framing/connection semantics must be followed rather than assuming every error can retain the connection. For application-level recoverable errors, use an authorized private error destination. Use standard close codes such as 1008 for policy violation, 1009 for oversized messages and 1011 for server failure as applicable; do not leak exception details. Define resynchronization after disconnect.

Figure 7. Attack points and the controls at each boundary

Security boundaries: attack points and practical defenses

The BFF reduces browser token theft; each remaining boundary still needs its own controls.

A · Browser / MFE

XSS or compromised remote code

CSP, safe output, controlled releases
No OAuth tokens in JavaScript

B · Cookie session

Forged change request / CSRF

Secure + HttpOnly cookie; CSRF token
Origin validation and bounded lifetime

C · Edge / callback

Spoofing, overload or code leakage

WAF, route limits, trusted forwarding
State / nonce / PKCE; safe logging

D · BFF / shared store

Credential theft or arbitrary proxying

Private encrypted store, ACLs, rotation
Outbound host and path allowlists

E · Business API

Unauthorized record or field access

JWT validation + current RBAC / ABAC
Trusted resource scope; safe DTOs

F · Data / audit

Injection, bulk export or tampering

Parameterized SQL, restricted DB roles
Bounded exports; durable protected audit

G · Live WebSocket

Stale or unauthorized event delivery

Authorize subscribe and delivery
Logout fan-out; bounded expiry checks

H · Partner integration

Stolen machine credential / overreach

Dedicated client, scopes, quotas
Rotation; resource checks in API

Acceptance evidence: cross-site writes denied, wrong-audience tokens rejected, cross-terminal data blocked, logout races fenced, socket entitlements revoked, and retained logs free of credentials and callback codes.

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Illustrative topology · proposed controls

14. Code structure, OpenAPI and quality gates

Reference Java package boundaries:

TEXT · illustrative specification

```
com.aobp.flight
  api              # controllers, request/response DTOs, exception advice
  application      # use cases and transaction orchestration
  domain           # flight rules, domain types and typed exceptions
  infrastructure   # repositories, outbox, external adapters
  security         # authorization facade and policy adapter
  observability    # schema adapter and business metrics
```

Keep the gateway reactive and independent; BFF can use the Spring web stack selected for session/WebSocket needs; Flight API can use Spring MVC/JPA. Do not add both servlet and reactive web starters casually. Shared dependencies supply contracts, error factories and telemetry conventions; they must not contain flight domain logic or centralize all services in one release cycle.

Maintain an OpenAPI document per BFF/domain API and AsyncAPI or equivalent schemas for event/WebSocket contracts. Pin a supported OpenAPI version in tooling (3.1 is the reference if all generators/validators support it). Document security, operationId, permissions, query bounds, DTOs, nullability, headers, all status codes, Problem Details and examples. Gate breaking changes in CI and generate TypeScript clients after contract approval. JWT bearer security applies to domain APIs; BFF declares cookie-session security and documents CSRF, with custom UI handling where tooling lacks support. Publishing OpenAPI does not authorize operations.

Example operationId names: listFlights, getFlight, updateFlightOperationalTimes, createFlightExport. Specify read-only fields and additionalProperties rules in schemas. Unknown input fields are rejected for security-sensitive commands; clients tolerate additive response properties. Enum additions may break exhaustive clients, so classify them explicitly in compatibility rules. Do not change units, timestamp meaning, default sort or authorization semantics silently.

Gate	Evidence
URI/schema lint	Naming, version, operationId and examples conform
Contract compatibility	No unapproved breaking change; generated client integration passes
Negative API tests	Cross-terminal/object/field access denied; no mass assignment
Error consistency	Parse/security/gateway/domain failures follow correct contract
Concurrency	Two updates with one ETag cannot silently overwrite
Idempotency	Duplicate concurrent POST does not execute twice
Telemetry	Trace continuity, requestId and sanitized logs verified
WebSocket	Unauthorized subscribe/delivery, expiry, reconnect and fan-out tested
Capacity/reliability	Representative throughput, tail latency, timeout and outage behaviour measured
Deployment	Helm schema/security checks, migration compatibility and rollback verified

Definition of done: OpenAPI updated, contract and negative tests passed, policies approved, bounded dependencies configured, dashboards/alerts/runbook provisioned, audit delivery verified, image scanned/signed with SBOM, and deployment/recovery evidence retained. No claim of enterprise readiness is based solely on using a named product or pattern.

15. Standards reference register

Reference	Applied purpose
RFC 9700 OAuth Security BCP	OAuth threat model and protocol hardening
RFC 10017 Browser-Based Applications	BFF pattern and browser threat analysis
RFC 9110 HTTP Semantics	Methods, status, conditional requests and headers
RFC 9457 Problem Details	Standard HTTP API error representation
RFC 3339 timestamps	Interoperable date/time strings
RFC 7396 JSON Merge Patch	Documented partial-update representation
RFC 6585	Additional statuses including 428 and 429
W3C Trace Context	Cross-service trace propagation
OpenTelemetry Logs Data Model	Trace-correlated structured telemetry
OpenAPI	Machine-readable synchronous API contract
CloudEvents / AsyncAPI	Optional governed event/async contracts

Official sources: https://www.rfc-editor.org/rfc/rfc9110 ; https://www.rfc-editor.org/rfc/rfc9457 ; https://www.rfc-editor.org/rfc/rfc3339 ; https://www.rfc-editor.org/rfc/rfc7396 ; https://www.rfc-editor.org/rfc/rfc6585 ; https://www.w3.org/TR/trace-context/ ; https://opentelemetry.io/docs/specs/otel/logs/data-model/ ; https://spec.openapis.org/oas/ ; https://cloudevents.io/ . These standards guide the design; AOBP-specific choices are explicitly identified above.

16. Technology stack

Capability	Reference stack	Selection note
UI	React + TypeScript, independently built shell/remotes	Federation implementation depends on bundler
Static hosting	Nginx or equivalent static service	Separate image per frontend
Gateway	Spring Cloud Gateway Server WebFlux	Replaceable through standard contracts
BFF/API	Supported Spring Boot + Spring Security	Separate services and dependency sets
Identity	Keycloak + dedicated PostgreSQL	Corporate OIDC/SAML federation
Sessions	Redis-compatible store	Verify client compatibility, support and exact selected distribution's licence
Flight state/audit outbox	PostgreSQL	Independent roles and schemas; audit archive separate
Business events	Kafka or RabbitMQ	Choose one based on replay/fan-out needs; not both by default
Policy	Spring classes; OPA adapter where shared governance needed	Version and test policies
Metrics	Micrometer, Prometheus, Alertmanager	Private scrape endpoints
Traces	OpenTelemetry Collector + Tempo	Alternative Jaeger possible
Logs	Structured JSON, collector + Loki	OpenSearch alternative for richer search
Dashboard	Grafana	OIDC login, scoped operational access
Packaging	Docker, Compose for development; Helm for Kubernetes	Immutable images
Delivery	CI plus Argo CD or equivalent GitOps	Build once, digest promotion

Open source is not the same as permissive licensing for every component. Review pinned component/image/plugin licences, including Grafana/Loki/Tempo distribution choices and the selected session store. Community software does not include a contractual support SLA. Produce an SBOM and licence inventory during delivery.

17. Complete request tracing and error tracking

Telemetry path: browser diagnostics pass through an authenticated, rate-limited BFF ingestion endpoint to the collector. Gateway/BFF/API traces go to OpenTelemetry Collector → Tempo; structured logs go to Loki; private service metrics are scraped by Prometheus. Grafana joins these views and Alertmanager routes operational alerts.

Use one documented instrumentation strategy per service. Avoid simultaneously enabling overlapping agents and Spring instrumentation that create duplicate server spans. Trace ingress/gateway, BFF and API separately and propagate `traceparent`/`tracestate` over approved boundaries. Enable sampling, exporters and log correlation explicitly; propagation alone does not export spans.

Browser SDK creates spans for navigation, remote load and approved API calls, tags MFE name/version, and reports sanitized JS errors/unhandled rejections and performance measurements. It is an implementation task, not an automatic capability of React. Initial navigation can precede SDK initialization; use navigation/resource timing rather than promising one browser trace covers every bootstrap request.

Browser telemetry credentials must not be secrets embedded in JS. Expose a restricted ingestion proxy with origin checks, rate/body limits, schema filtering and abuse controls. Do not expose a generic Collector receiver publicly. Propagate traces only to allowlisted destinations. Scrub query strings, user input and source-map payloads; protect private source maps used for symbolication. Treat browser-provided trace identifiers as untrusted correlation data, never authorization.

Structured fields: timestamp UTC, severity, service.name, service.version, deployment.environment, site, namespace, pod, route_id, MFE version where known, trace_id, span_id, request_id, event_id, workflow_instance_id, flight resource reference where permitted, error.code, outcome and duration. Avoid personal details and secrets. Gateway owns a sanitized request ID; client correlation values are length/format checked. No tokens/cookies/bodies by default.

HTTP spans capture transport status; domain events capture business outcomes. A 200 response can contain a business rejection and a 403 is not necessarily a software defect. Standard error response: stable error code, safe message, request/trace reference; detailed stack traces remain private. Track error fingerprints to group repeated defects.

Async events carry trace metadata plus event ID, causation ID and correlation/business-process ID. For fan-out or long delays, use trace links/new consumer traces rather than one unbounded trace. Metrics use bounded labels: service, route, operation, outcome. Never label by user/flight/trace/request ID or raw URL.

Telemetry has a defined loss policy. Sample traces; never present sampled traces as complete audit evidence. Tail/error sampling requires collector capacity and correct trace routing. Maintain durable audit independently. Bounded collector queues and retry storage prevent observability outages from exhausting application memory; monitor dropped telemetry.

Browser-entry logging policy

The BFF telemetry endpoint validates schema/size and strips secrets before forwarding. Do not expose unrestricted public OTLP ingestion. The browser ALB carries OIDC callback codes in query parameters; this design disables access logging on that ALB until an approved capture/redaction design exists. Classic ALB access records contain the request URL; gateway redaction cannot sanitize a record already captured by ALB. Enforce `access_logs.s3.enabled=false` on the browser ALB in the reference configuration, and test any separately enabled enhanced logging destination before use. Use sanitized gateway/BFF access and security events for request audit. This is a deliberate visibility trade-off, not a claim that application redaction sanitizes ALB logs.

WAF logging and request sampling need separate controls: redact Authorization, cookies and sensitive query arguments, and disable sensitive request sampling or configure validated data protection. Partner ALB logs use a separately approved policy. Verify actual retained records during security acceptance; restrict access and retention on every logging destination.

Dashboard provisioning specification

Version-control data sources, dashboard JSON, alert rules and permissions. Use stable data-source UIDs (`prometheus`, `loki`, `tempo`). Configure trace-to-log links using trace_id; log-to-trace derived fields; metric exemplars where instrumentation and backend support them. Use dashboard variables environment/site/service/route/release. Do not expose broad data-source query access merely because a dashboard has filters.

Dashboard	Panels and drill-down
Platform overview	Service health, traffic, errors, p95/p99, deployment versions, saturation
Gateway	Per-route rate/latency/status, throttles, upstream failures, connections, event-loop pressure
BFF and identity	Session/login/refresh failures, session-store latency, revocation processing
Flight API	Endpoint latency, error codes, DB pool/waits, policy decisions, update conflicts
WebSocket	Active sessions, handshake failures, connection lifetime, messages/sec, delivery lag, disconnect reason, queue pressure
Browser/MFEs	Remote-load failures, JS errors, navigation/Web Vitals, API failures by release
Business operations	Flight-event freshness, overdue tasks, rejected updates, integration backlog
Audit/security	Denied actions, privilege/config changes and audit outbox age

WebSocket connection duration is not message latency. Add application metrics/spans for message processing and delivery. In a microfrontend page, Web Vitals are page-level; attribute remote actions/errors separately rather than claiming every page metric belongs exclusively to one remote.

Starter PromQL templates (verify actual exported metric/label names):

PROMQL · illustrative specification

```
sum by (routeId) (rate(spring_cloud_gateway_requests_seconds_count[5m]))
histogram_quantile(0.95,
  sum by (le, routeId) (rate(spring_cloud_gateway_requests_seconds_bucket[5m])))
sum(rate(http_server_requests_seconds_count{service="flight-api",status=~"5.."}[5m]))
/
```

```
clamp_min(sum(rate(http_server_requests_seconds_count{service="flight-api"}[5m])), 0.001)
```

Histogram buckets must be enabled. Prometheus scrape labels such as service must be configured consistently. Set low-traffic conditions for ratio alerts. Example triggers: sustained elevated 5xx with minimum traffic; p95 above agreed SLO; zero ready replicas; oldest outbox event over agreed threshold; rising event lag; dropped telemetry; certificate expiry; authorization service unavailable. Numerical thresholds are open decisions based on the workload. Route alerts to the correct support owner, with dashboard/log/trace links and runbooks.

18. Kubernetes deployment and Helm specification

EKS deployment overview

Figure 5. AWS EKS placement, routing and private data tier
See the landscape diagram plate following this section.

The reference AWS footprint spans three Availability Zones where available. Figure 5 shows logical tiers across zones; it does not require a gateway, BFF and API request to stay within one zone. ALB uses at least two eligible subnets in different zones. Run worker pods in private subnets without public IPs. A private/internal ALB is the default for BAC staff access over approved corporate connectivity; use an internet-facing ALB only for an approved public audience. In either case, pod and database endpoints remain private. Internet-facing entry is not a prerequisite for SSO.

Route 53 supplies application and identity names; ACM supplies the external TLS certificate. AWS WAF is associated with approved ALBs. The Load Balancer Controller uses a least-privilege IAM role through an approved EKS workload-identity mechanism. EKS control-plane endpoint access is private/restricted. ECR, Secrets Manager, CloudWatch and other required services use VPC endpoints where practical; approved NAT/egress paths serve required external IdP endpoints.

Data plane versus control plane

The platform team creates Ingress and ClusterIP Services. The controller reads those objects and manages ALB listeners, target groups and pod registration. For target-type ip, the ALB forwards directly to selected pod IPs, rather than first traversing a worker NodePort. The Ingress object and the controller are configuration/control components, not two application proxies in the request path. This distinction avoids drawing or operating duplicate ingress layers.

Service and exposure specification

Workload	Kubernetes object / exposure	Starting availability policy
Gateway	Deployment + ClusterIP; ALB IP targets	Minimum 3 pods across zones; HPA; PDB minAvailable 2
BFF	Deployment + ClusterIP; gateway-only application ingress	Minimum 3 pods across zones; HPA; PDB minAvailable 2
Shell and each remote	Independent Deployment + ClusterIP; ALB targets	Minimum 2 pods; immutable static asset versions
Flight / Workflow APIs	Independent Deployments + ClusterIP	Minimum 2 pods per service; scale from tested workload
Central Security Service	Deployment + ClusterIP	Minimum 2 pods; tested cache/freshness and fail-closed behavior
Keycloak	Supported HA deployment; isolated identity ingress	Follow selected release HA/database/cache guidance
Session store	Managed private Redis-compatible replication group	Multi-AZ failover; TLS/ACL/encryption; capacity and loss behavior tested
Business / authorization / identity databases	Private PostgreSQL services, distinct ownership/grants	Multi-AZ managed DB where selected; backup/PITR and restore tests

Replica counts are initial resilience proposals, not capacity estimates or guaranteed SLOs. Three pods do not guarantee three-zone placement without topology rules and available nodes. A PDB constrains voluntary disruptions; it does not prevent failures or replace surge capacity. Test node/AZ loss and managed-store failover. Session loss may require reauthentication rather than loss of business data.

What actually distributes a request

ALB distributes each new HTTP request among healthy registered gateway targets. The gateway resolves the BFF ClusterIP Service, which provides access to eligible BFF endpoints using the cluster's supported networking implementation. BFF resolves the fixed private API Service. Kubernetes handles pod replacement and endpoint updates; it does not authenticate the airport operator.

The three BFF replicas are interchangeable for HTTP because session, login-correlation and authorized-client state are shared. They do not each hold an independent login for the same browser. A WebSocket stays on the accepting pod until it closes; a reconnect is a new authorized handshake and can land on another pod.

Namespace and network boundaries

Use aobp-edge, aobp-apps, aobp-identity and aobp-observability. Namespace names do not enforce isolation. Use a NetworkPolicy-capable CNI, default-deny policies, security groups and approved egress. Evaluate security groups for pods where useful; document their interaction with CNI network-policy mode and ALB source traffic.

Allowed source	Allowed destination	Purpose
Approved client network	ALB 443	Browser/approved partner entry

Allowed source	Allowed destination	Purpose
ALB security group	Static or gateway TLS application port	Only registered application targets and health checks
Gateway pods	BFF application TLS port	Browser dynamic routes; no BFF management API
Gateway pods	Approved business API TLS ports	Partner paths only
BFF pods	Flight/Workflow APIs, session store, Keycloak	Token relay, clustered session and OAuth back channel
APIs	Required policy service, own DB, broker	Authorized business execution
Telemetry agents	Private metrics/OTLP endpoints	Controlled collection
Each workload	Approved DNS and explicit infrastructure endpoints	Resolution, secrets and required service APIs

Health-check rules are explicit exceptions. Give ALB a minimal 200-only readiness endpoint on the application listener, without build details or management controls. Management endpoints on a separate port remain available only to authorized probes/collectors/operators. A compromised frontend pod must not be able to reach Redis, databases or private APIs.

TLS and trusted forwarding

Terminate external HTTPS at ALB and use HTTPS target groups for gateway and static workloads. Use certificate-verifying TLS for gateway-to-BFF and BFF-to-services; mTLS may be implemented with Istio or another supported workload-identity system. A mesh is optional, and never replaces JWT/business authorization.

ALB HTTPS target connections encrypt traffic but ALB does not validate target certificates. Do not describe this hop as certificate-verified mutual authentication. Security groups and target registration constrain it. If policy requires end-to-end authenticated transport from the external boundary, assess a validated proxy/mTLS termination design or NLB TLS passthrough with the required WAF/routing trade-offs. This is a specific security decision, not an undocumented switch in a Helm chart.

Configure trusted proxies so spoofed client Forwarded/X-Forwarded headers cannot affect redirect URLs, secure-cookie decisions or audit IPs. Validate public issuer and callback consistency; do not change Keycloak's issuer to an internal service DNS name. Back-channel connectivity may use split-horizon DNS provided certificate identity and issuer remain consistent.

Ingress implementation template

This fragment illustrates only the BFF-related ALB rules and references an already defined gateway Service in the same namespace. Supply real certificate/WAF identifiers, subnets, health checks, static-host rules and security groups through controlled environment configuration. It is not a complete deployment package.

YAML · illustrative specification

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: aobp-browser-dynamic
  namespace: aobp-edge
```

```
  annotations:
    alb.ingress.kubernetes.io/scheme: internal
    alb.ingress.kubernetes.io/target-type: ip
    alb.ingress.kubernetes.io/listen-ports: '[{"HTTPS":443}]'
    alb.ingress.kubernetes.io/backend-protocol: HTTPS
    alb.ingress.kubernetes.io/healthcheck-protocol: HTTPS
    alb.ingress.kubernetes.io/healthcheck-path: /health/ready
    alb.ingress.kubernetes.io/success-codes: '200'
    alb.ingress.kubernetes.io/load-balancer-attributes: >-
      idle_timeout.timeout_seconds=120,access_logs.s3.enabled=false,
      routing.http.drop_invalid_header_fields.enabled=true
spec:
  ingressClassName: alb
  rules:
    - host: aobp.example.com
      http:
        paths:
          - path: /bff
            pathType: Prefix
            backend:
              service:
                name: aobp-gateway
                port:
                  name: https
          - path: /oauth2/authorization
            pathType: Prefix
            backend:
              service:
                name: aobp-gateway
                port:
                  name: https
          - path: /login/oauth2/code
            pathType: Prefix
            backend:
              service:
                name: aobp-gateway
                port:
                  name: https
```

Use explicit TLS policies and ACM certificate configuration in the full Ingress. With a separate Ingress per remote/namespace, centrally govern any shared IngressGroup: it is a trust boundary because members can affect listener rules. Limit RBAC/admission permissions, assign rule priorities and test the shell catch-all. Do not let an unrelated team join a shared group and override authentication routes. A shared ALB may have separate gateway target groups by host/route; partner and browser gateway Deployments may be split for stronger isolation.

Pod lifecycle and resource specification

Chart defaults include non-root UID, read-only root filesystem with explicit writable tmp volume, dropped capabilities, RuntimeDefault seccomp, resource requests, startup/readiness/liveness probes and graceful shutdown. Disable service-account token automount when the application does not need the Kubernetes API; use a dedicated least-privilege AWS identity only for required AWS APIs.

Start BFF/gateway sizing experiments around 500m CPU and 1 GiB memory requests per pod with a tested memory ceiling, then measure representative load. Include JVM/native/TLS/connection overhead. HPA uses CPU plus measured concurrent requests or connection pressure as appropriate. WebSocket load and Redis latency can saturate before CPU. Pod count is not a substitute for queue and connection limits.

Configure topologySpreadConstraints, pod anti-affinity where appropriate, HPA minimums, node capacity and controlled disruption. Omit Deployment replicas when HPA owns it. Coordinate readiness removal, target deregistration, preStop and termination grace; test interruption with ongoing HTTP/WS traffic. A proposed ALB idle timeout of 120 seconds requires protocol heartbeats below that interval.

Keep authentication/session expiry independent of transport heartbeats. Bound socket drain and use reconnect plus authorized snapshot recovery.

Liveness checks process health; it must not restart all pods because Redis or Keycloak is down. Readiness reflects safe serving capacity and required dependencies without turning optional telemetry outages into platform-wide failure. Expose separate management ports only through private policies.

Helm and delivery structure

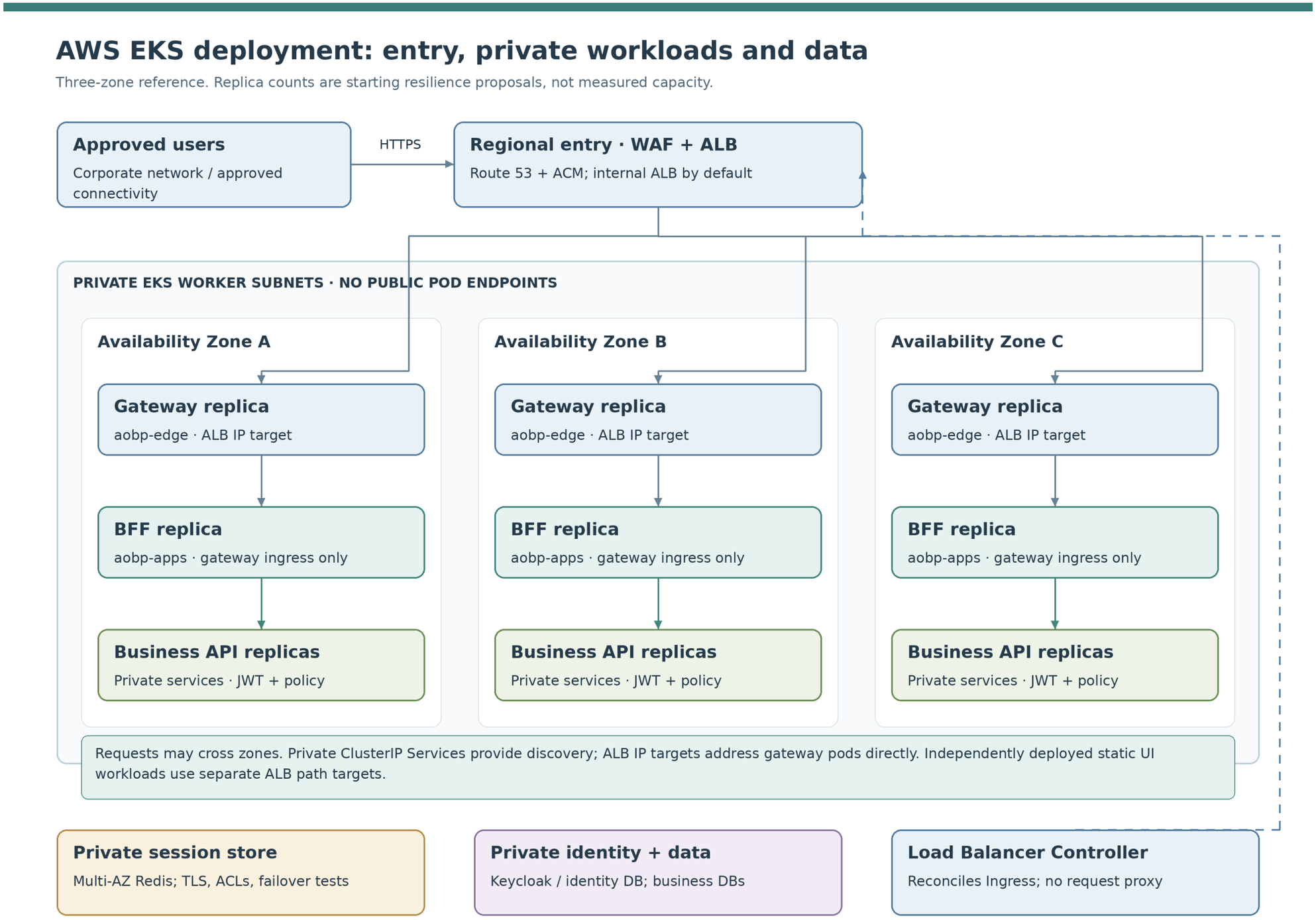
TEXT · illustrative specification

```
charts/
  aobp-edge/                # gateway and managed entry configuration
  aobp-shell/               # independent shell assets
  aobp-flight-mfe/          # independent flight remote
  aobp-workflow-mfe/        # independent workflow remote
  aobp-bff/                 # session, OAuth, REST and WebSocket
  aobp-flight-api/          # business API and policy adapter
  aobp-security/            # policy runtime/configuration
  aobp-observability-config/ # dashboards, collectors and alerts
environments/
  dev/ staging/ prod-onprem/ dr-aws/
```

Use values.schema.json and validated image digests; pin charts and platform versions. Secrets use approved Secrets Manager integration with workload-scoped IAM; never Git/ConfigMap/plain Helm values. Rotate BFF client credentials, encryption keys and database passwords through tested procedures. Use backward-compatible database migrations, controlled migration jobs and expand/contract rollout. Helm rollback does not undo a schema migration. Set total connection pools against maximum replicas and database limits.

CI checks include contracts, negative security tests, secret/dependency/image/IaC scans, SBOM/signing, helm lint/template, Kubernetes schema validation, admission policy and realistic login/refresh/WS tests. Argo CD or equivalent applies reviewed environment changes. Promote the same digests and retain restore/rollback evidence. No second load balancer is created for each private BFF/API Service.

Figure 5. AWS EKS placement, routing and private data tier



19. Docker and Docker Compose execution design

Each backend produces one non-root image. Build Java artifacts in a builder stage with pinned toolchain; copy only the artifact into a supported pinned JRE runtime. Frontends use a Node builder and a static web runtime; no frontend dev server in production. Supply frontend environment/remote manifest at runtime using an external configuration endpoint/file, not baked secrets.

Dockerfile outline (replace placeholders with selected pinned images/artifact):

DOCKERFILE · illustrative specification

```
FROM company-approved-jre@sha256:REPLACE AS runtime
WORKDIR /app
COPY --chown=10001:10001 app.jar /app/app.jar
USER 10001:10001
EXPOSE 8080 9090
ENTRYPOINT ["java", "-jar", "/app/app.jar"]
```

Compose is for local integration and single-host demonstration, not production HA. Use the same images and external route contracts as Kubernetes. A local edge proxy routes static hosts/paths, gateway and identity without coupling UI images to gateway. Set local DNS/hosts entries, trusted development TLS and reachable OIDC issuer names first. Browser and containers must resolve the same configured issuer correctly; `localhost` inside a container is not the host or Keycloak.

Service inventory:

Compose service	Dependencies
edge	shell, remotes, gateway, keycloak
shell, flight-mfe, workflow-mfe	Their independent static images
gateway	BFF; private management metrics
bff	keycloak, sessions/authorized-client store, flight-api, event broker
flight-api	flight-db, policy if external, event broker
keycloak	identity-db
outbox-relay	flight-db and event broker
prometheus, grafana, collector, tempo, loki	Provisioned monitoring configuration

Compose skeleton illustrating contracts (not a runnable complete file):

YAML · illustrative specification

```
services:
  gateway:
    image: ${GATEWAY_IMAGE:?Set an immutable image reference}
    env_file: ./config/gateway.env
    networks: [app]
    restart: unless-stopped
  bff:
    image: ${BFF_IMAGE:?Set an immutable image reference}
    env_file: ./config/bff.env
```

```
    secrets: [bff_oidc_secret]
    networks: [app, data]
    restart: unless-stopped
  flight-api:
    image: ${FLIGHT_API_IMAGE:?Set an immutable image reference}
    env_file: ./config/flight-api.env
    secrets: [flight_db_password]
    networks: [app, data]
    restart: unless-stopped
  flight-mfe:
    image: ${FLIGHT_MFE_IMAGE:?Set an immutable image reference}
    networks: [app]
  grafana:
    image: ${GRAFANA_IMAGE:?Set a pinned approved image}
    volumes:
      - ./observability/grafana/provisioning:/etc/grafana/provisioning:ro
      - grafana-data:/var/lib/grafana
    networks: [app]
networks:
  app: {}
  data:
    internal: true
volumes:
  grafana-data: {}
secrets:
  bff_oidc_secret:
    file: ./local-secrets/bff_oidc_secret
  flight_db_password:
    file: ./local-secrets/flight_db_password
```

Applications must explicitly consume mounted secrets, e.g. via supported Spring configtree import and correctly named files. Compose secrets are file delivery, not a production secret manager. Keep local-secrets excluded from source control. Define healthchecks with tooling actually present in images; add health-based depends_on for startup convenience, but retain application retries because dependency failure occurs after startup too. Publish only local edge/required operator ports; keep DB/Redis/broker/admin interfaces private. Supply persistent volumes and backup/reset commands. Never use `down -v` when data must be retained.

Execution contract after implementing the complete package:

BASH · illustrative specification

```
docker compose --env-file .env.local config --quiet
docker compose --env-file .env.local up -d
docker compose --env-file .env.local ps
docker compose --env-file .env.local logs -f gateway bff flight-api
docker compose --env-file .env.local down
```

Kubernetes execution contract after charts, dependencies and secret provisioning are complete:

BASH · illustrative specification

```
helm lint charts/aobp-flight-api
helm template flight-api charts/aobp-flight-api -f environments/dev/flight-api.yaml
helm upgrade --install flight-api charts/aobp-flight-api --namespace aobp-apps --create-namespace -f
    environments/dev/flight-api.yaml --wait
```

20. Data, audit and resilience

Flight API owns flight state and trusted object attributes. Keycloak owns identity state in an independently protected database. Sessions are transient but contain sensitive material. Business history, security audit and diagnostic telemetry have separate retention/access policies.

Transactionally persist flight changes plus outbox records. Consumers deduplicate, observe event ordering/resource versions and reconcile after downtime. Capture denied requests and failed authentication through a separate security-event path. Protected immutable archive provides stronger audit assurance than an append-only SQL table alone.

On observability outage, bounded buffers and a documented loss policy protect business operations. On audit persistence failure, critical audited mutations fail according to approved business policy. On policy outage, fail closed for protected actions. On session-store outage, return controlled unavailable responses rather than creating anonymous/partial sessions. On IdP outage, new login fails; existing sessions follow expiry/refresh rules with no hidden local-password fallback.

On-prem/AWS DR: independent entry and gateway capability, same approved application/configuration digests, replicated/backed-up business/identity data, recoverable signing keys/secrets, explicit fencing of primary writers, DNS/routing cutover and partner routing. Do not stretch a session/quota/database quorum across sites casually. Prefer an explicitly tested active/passive design until latency and consistency requirements justify more. Reauthentication after failover is an acceptable proposed simplification, subject to business agreement. Recovery targets remain to be approved per service.

21. Acceptance and implementation sequence

1. Establish external routes, DNS/TLS, issuer, permission catalogue and OpenAPI contracts.

2. Deliver shell plus one independent flight remote; demonstrate remote-only deployment and rollback.

3. Implement login/BFF session/Flight API authorization with negative cross-terminal tests.

4. Add authorized WebSocket updates and multi-replica event delivery, logout/expiry and reconnect recovery.

5. Provision observability, dashboards and alerts; demonstrate browser error to API trace/log drill-down and metric exemplar where supported.

6. Implement Docker Compose integration environment and independent Helm releases.

7. Run capacity, dependency outage, security and DR exercises before approving production sizing.

Acceptance scenario	Required evidence
Independent remote deployment	Shell/gateway/BFF unchanged; old/new browser compatibility shown
Gateway replacement	Login callbacks, cookies, REST, WS and tracing contracts pass
Cross-terminal request/subscription/export	Denial at backend and no event/data leakage
Account disabled with open socket	Access removed within agreed revocation bound
BFF replica terminated	Client reconnects and recovers correct authorized snapshot
Multi-replica event delivery	Authorized users on different BFFs receive updates
Token/session inspection	No OAuth credentials in browser storage/responses; opaque cookie flags verified; no secrets in logs/telemetry
Tracing drill-down	Request connects gateway/BFF/API spans and relevant logs
Slow DB/event broker	Bounded resources, useful error and operator alert
Telemetry backend outage	Business service remains bounded; loss/backlog visible
Outbox replay	No duplicate business mutation; history reconciles
DR exercise	Fenced writes, identity recovery and audited cutover/failback

Security-review acceptance evidence

Scenario	Pass condition
Inspect browser traffic/storage	No API access/refresh/ID token in React storage, JS responses or URLs; session cookie is opaque/HttpOnly
Complete login on a different BFF pod	Shared pre-login correlation validates successfully without stickiness
Refresh concurrently on multiple pods	One safe refresh outcome; no stale overwrite or rotating-token replay loop
Forge callback state or return URL	Rejected; no open redirect or session fixation
Cross-site mutation / logout attempt	CSRF rejected and no downstream action occurs
Spoof X-User, X-Roles, Host or bearer header	No identity substitution or redirect poisoning
Call business API directly from browser route	No bypass of BFF; private services inaccessible
Present wrong-audience/expired/ID token	API rejects before business execution
Terminate BFF or fail over Redis	Controlled recovery/error; no anonymous fallback or credential leakage
Expire token across multi-service call	No blind duplicate mutation; safe reconciliation and tested error handling
Disable user with open WS	REST and event delivery revoked within the approved bound
Logout races with refresh	In-flight refresh cannot recreate session/token state after revocation or absolute expiry
Back-channel logout callback	Valid IdP logout closes mapped sessions; replayed or forged logout tokens fail
Probe logs and edge logging configuration	Browser ALB raw access logging is disabled; retained application/WAF records contain no credentials, callback code/state or session identifiers
Deploy policy/store outage	Protected operations fail closed with controlled 503, not restart storms

Production sign-off requires results, not only the presence of these controls in configuration.

22. Open decisions and provisional register

Selected architecture decisions

ID	Decision	Advantage and trade-off
ADR-01	BFF owns web OAuth; React receives session cookie only	Reduces browser token exposure; adds a sensitive server component
ADR-02	ALB → gateway → BFF for dynamic browser traffic	Uniform traffic protection before session logic; adds one application hop
ADR-03	Gateway and BFF are separate Deployments	Independent ownership/scaling; requires compatible route and timeout contracts
ADR-04	BFF calls private services directly	Avoids public hairpin and duplicate edge checks; services must enforce their own security
ADR-05	ALB IP targets; no additional ingress-controller proxy	Clear EKS path; depends on supported networking/controller configuration
ADR-06	Same external origin, independent frontend releases	Simple cookie/CSRF integration; MFEs still share JavaScript privileges
ADR-07	Shared session and authorized-client store with coordinated refresh	Requests survive pod changes; creates a critical sensitive shared dependency
ADR-08	Partner bearer routes bypass human BFF	Correct machine identity model; strict route/client isolation is required
ADR-09	Authoritative RBAC/ABAC in business services	Protects object/field scope; policy freshness and availability need ownership
ADR-10	Single aobp-api audience initially	Simple common API boundary; split audiences for materially separate trust domains

These are the recommended design decisions for this revision. Environment sizing, operational policy and formal implementation approvals remain separate.

Values and choices to finalize before production

Item	Proposed baseline	Owner and required evidence
EKS deployment role	AWS runtime for production or secondary/DR	Platform/continuity leads confirm environment role
Entry audience	Internal ALB for staff; public only where approved	Network/security approve client connectivity and SSO reachability
Identity integration	Keycloak broker to BAC IdP	IAM validates federation, group overage, logout and metadata policy
Session lifetime	30-minute idle; 8-hour absolute	Security/business approve and test UX/lifecycle
Access-token lifetime	5 minutes; 60-second refresh margin	IAM validates actual issuance, rotation and concurrency behavior
Disablement propagation	Target 60 seconds for protected actions	Service/policy owners prove fresh revocation/entitlement enforcement
Runtime sizing	3 gateway/BFF replicas; measured per-pod budgets	Platform team completes representative load and AZ-loss tests
Session store	Managed private Redis-compatible service	Platform team validates client compatibility, failover, licensing and encryption
Broker fan-out	Kafka or RabbitMQ, one selected deliberately	Integration team proves delivery to all relevant BFF replicas
Transport trust	HTTPS plus network restrictions; verified service TLS	Security approves ALB target certificate limitation and mesh choice
Recovery	On-prem primary / AWS active-passive direction	Continuity team approves RTO/RPO, fencing and tested failback

Implementation responsibility matrix

Owner	Concrete deliverable	Acceptance evidence
IAM team	BFF OIDC client, corporate federation, machine clients, token/logout configuration	Login, audience, MFA, rotation and back-channel logout tests
Platform team	ALB/WAF rules, private EKS services, Redis, secrets and network policies	Allowed/denied network tests, TLS review, pod/AZ/store-failure evidence
BFF team	Session/correlation storage, CSRF, allowlisted relay, coordinated refresh and session invalidation	Multi-pod callbacks, concurrent refresh and browser credential inspection
API teams	JWT validation, object/field authorization, idempotency and durable audit	Wrong-token rejection, cross-terminal denial and safe retry tests
UI team	Relative BFF calls, safe login/logout UX, CSRF headers and governed MFEs	No OAuth token in browser storage/responses; safe 401/403/503 handling
Operations/security	Dashboards, credential-safe logging, revocation/DR runbooks	Trace drill-down, retained-log inspection and timed recovery exercise

Alternatives considered

Direct SPA OAuth remains a legitimate pattern for some applications but does not satisfy this review's server-side token boundary. Combining BFF and gateway can reduce deployable count, but couples session security and edge traffic management; separate services fit AOBP's shared platform and independent deployment model. Placing a single internal gateway after BFF does not by itself protect incoming BFF traffic and would require another entry design. Adding gateways on both sides of BFF is unnecessary until a clear internal governance requirement exists. A token-mediating backend that gives access tokens back to React is not the selected BFF architecture.

23. Official implementation references

These are implementation references, not evidence of deployed AOBP capabilities. Verify documentation against the pinned release.

- Spring Cloud Gateway: https://docs.spring.io/spring-cloud-gateway/reference/
- Gateway Actuator API: https://docs.spring.io/spring-cloud-gateway/reference/spring-cloud-gateway-server-webflux/actuator-api.html
- Spring Security OAuth2 client: https://docs.spring.io/spring-security/reference/servlet/oauth2/client/index.html
- Spring Security Resource Server JWT: https://docs.spring.io/spring-security/reference/servlet/oauth2/resource-server/jwt.html
- Spring Security WebSocket: https://docs.spring.io/spring-security/reference/servlet/integrations/websocket.html
- Spring Session WebSocket: https://docs.spring.io/spring-session/reference/guides/boot-websocket.html
- Keycloak production: https://www.keycloak.org/server/configuration-production
- Keycloak HA: https://www.keycloak.org/high-availability/introduction
- OpenTelemetry sensitive data: https://opentelemetry.io/docs/security/handling-sensitive-data/
- OPA external data: https://www.openpolicyagent.org/docs/external-data
- Kubernetes documentation: https://kubernetes.io/docs/
- Helm chart best practices: https://helm.sh/docs/chart_best_practices/
- Docker Compose: https://docs.docker.com/compose/
- Grafana documentation: https://grafana.com/docs/grafana/latest/

Reference application and review notes

The selected BFF boundary follows RFC 10017 section 6.1; Authorization Code protection follows RFC 9700. AWS documentation explains the ALB IP-target/controller distinction and target TLS limitation. Spring documentation explains the authorized-client, JWT and CSRF mechanisms. None of these sources mandates the precise ALB → gateway → BFF ordering, the proposed replica counts or AOBP's session lifetimes. Those choices are recorded as local decisions and must be tested with the selected product versions.

Additional sources for this revision

Reviewed 2 November 2026. The references define mechanisms and guidance; the chosen ordering, replica counts and policy values are AOBP design decisions.

- [R1] RFC 10017, OAuth 2.0 for Browser-Based Applications, August 2026, section 6.1: https://www.rfc-editor.org/rfc/rfc10017.html
- [R2] RFC 9700, OAuth 2.0 Security Best Current Practice: https://www.rfc-editor.org/rfc/rfc9700.html
- [R3] AWS EKS, ALB ingress: https://docs.aws.amazon.com/eks/latest/userguide/alb-ingress.html

- [R4] AWS EKS best practices, load balancing and IP targets: https://docs.aws.amazon.com/eks/latest/best-practices/load-balancing.html
- [R5] AWS Load Balancer Controller, ingress annotations: https://kubernetes-sigs.github.io/aws-load-balancer-controller/latest/guide/ingress/annotations/
- [R6] ALB target-group TLS behavior: https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-target-groups.html
- [R7] Spring Security, OAuth2 authorized-client management: https://docs.spring.io/spring-security/reference/servlet/oauth2/client/authorized-clients.html
- [R8] Spring Security, JWT resource-server configuration: https://docs.spring.io/spring-security/reference/servlet/oauth2/resource-server/jwt.html
- [R9] Spring Security, CSRF and SPA integration: https://docs.spring.io/spring-security/reference/servlet/exploits/csrf.html
- [R10] Keycloak reverse-proxy and hostname considerations: https://www.keycloak.org/server/reverseproxy

R1 supports the server-side OAuth boundary and browser threat discussion. R3-R5 support the EKS ALB/controller/IP-target mechanics. R6 qualifies the ALB target TLS trust statement. R7-R9 describe the framework mechanisms to implement and test. Existing engineering conventions remain AOBP contracts rather than claims of universal industry standards.

- [R11] AWS ALB access-log fields and logging behavior: https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-access-logs.html
- [R12] OpenID Connect Back-Channel Logout 1.0: https://openid.net/specs/openid-connect-backchannel-1_0.html

Revision 2.3 review record

This revision retains all 23 original section headings and existing API, logging, observability, packaging and resilience contracts. It adds a reader guide, an explicit gateway-placement comparison, fixed BFF-to-API mapping, a plain-language click-through example, a failure/UX table, the back-channel logout route and receiver contract, an EKS distribution explanation and named implementation owners. Seven diagrams use consistent typography and restrained blue, teal, green, purple and sand colors. Figures 1-5 replace the earlier illustrations; Figures 6-7 add partner integration and attack-point views. Their numbers match the relevant section references.

The supplied attachment could not be mounted; the saved document was recovered. A later version 4 was then reloaded, and its additional destination mappings, examples, logout receiver details and implementation responsibilities were retained while the new security and diagram work was reapplied. Official OAuth, AWS and Spring sources were checked on 2 November 2026. Design values remain proposals until the acceptance evidence in section 21 is supplied. This document does not claim that code or infrastructure has been deployed or tested.